A Computational Approach to Discovering Emergent Hate Speech on Social Media

Patrick Y. Wu, Venkata Dhanush Kikkisetti, Suneela Maddineni, and Nathalie Japkowicz
American University

Correspondence concerning this article should be addressed to Patrick Y. Wu, American University.
Email: patrickwu@american.edu

Abstract

The growth of social media has enabled unprecedented global connectivity and access to information, but social media platforms have also become channels for hate speech that targets individuals and groups based on identity and protected characteristics. While advances in machine learning and natural language processing have improved hate speech detection, these models are static and cannot be updated easily in an environment where terminology evolves continuously. Although manual inspection can identify emerging terminology, the volume of posts often proves prohibitive. We propose a novel computational methodology and app, the Hate Mitigation App (HMApp), that discovers emerging coded or non-coded hate speech terminology on social media. Specifically, the approach uses a set of researcher-selected anchor hate terms to identify new terminology associated with a specific type of hatred, such as antisemitism. We use this approach with five different types of hatred: antisemitism, anti-Black hatred, Islamophobia, anti-Asian hatred, and anti-LGBTQ+ hatred. Using two categories of evaluation metrics—one comparing discovered terms against existing lexicons and another validating them through online searches—we find that the approach identifies both established and novel terminology. Our findings underscore the potential of computational approaches to augment manual detection efforts and keep pace with the rapidly evolving nature of online hate speech. Though related to both, our approach is neither Hate Speech Detection nor Automated Terminology Extraction, which are often construed as classification tasks. Our method, in contrast, uses unsupervised learning, thus engaging in a discovery rather than a classification task, yielding ramifications for both our design and testing methodologies.

Keywords: computational methods, hate speech, hate terminology, social media, unsupervised learning

Introduction

We propose a novel computational methodology that aims to discover emerging coded and non-coded hate speech terminology on social media. Though related to both, our approach falls outside the realm of Hate Speech Detection (HSD) and Automated Terminology Extraction (ATE) methods. Rather, it uses a novel combination of existing technologies from both fields, along with a novel search criterion, to generate a solution to a task that had not previously been formulated: New Hate Speech Terminology Discovery (NHSTD). Furthermore, in contrast with most HSD and ATE methods, our approach relies on a purely unsupervised methodology, thus requiring no labeled data—a resource difficult and expensive to obtain—beyond the initial formulation of appropriate anchor terms for the particular type of hatred under consideration. Our work makes both research and practical contributions. On the research side, it adds to the research landscape by defining a discovery task tucked between HSD and ATE. It then proposes an unsupervised method guided by a minimal amount of extrinsic knowledge, the anchor terms, to create a focused search aimed at solving the proposed new task. On the practical side, it offers a highly expandable tool that does not rely on annotated datasets to

discover emergent hate terminology. Rather than treating all forms of hatred simultaneously, thus losing some of the nuances necessary to understand them, it treats each separately. The unsupervised design and the modularity of our tool make it quite versatile, as it can easily adapt to different targets of hatred.

Our research originated from the observation that more than two in three people worldwide are now on some form of social media, and nearly 94% of the world's internet users use social media each month. While these platforms have enabled unprecedented global connectivity and democratized access to information, they have also become vectors for harmful content, including hate speech that targets individuals and groups based on protected characteristics such as race, religion, gender, and sexual orientation. The real-world consequences of online hate speech range from psychological harm to victims to the incitement of offline violence and harassment (Arcila Calderón et al., 2024; Saha et al., 2019; Williams, 2019). Given the exponential growth in user-generated content, manual moderation alone is insufficient to address hate speech at scale. At the same time, hate speech terminology evolves both endogenously, such as evolving within online discussions (Goel et al., 2023), and exogenously, such as in response to current events, making automated HSD tools capable of handling large volumes brittle (Olteanu et al., 2018).

Hate speech on social media has been steadily increasing over the years. According to the most recent annual report conducted by the Anti-Defamation League (2024), a survey-based report, severe incidents of online hatred and harassment increased from 18% in 2023 to 22% in 2024, while physical threats increased from 7% to 10%. The report noted a particularly sharp spike in harassment targeting disabled people, climbing from 35% in 2023 to 45% in 2024, with severe harassment jumping from 20% to 31%. Among all the marginalized groups surveyed, the LGBTQ+ community experienced the highest levels of severe online harassment, which surged from 30% to 45%. Physical threats against this community more than doubled from 6% to 14%.

Similarly, Madhu (2025) shows in a machine learning-based retrospective study of Twitter from 2015 to 2020 that the incidence of hate speech doubled during this period and was correlated with the platform's user base growth. Notably, this study was conducted prior to online platforms scaling back content moderation and major destabilizing events such as the January 6 Capitol attack and the October 7 Hamas attack.

Hate speech is not only increasing, but the terminology surrounding it is also rapidly evolving (Dubois & Reepschlager, 2024; Hoeken et al., 2023). Hate speech evolves in response to shifting current events, new content moderation systems, and the pressures of social desirability bias (e.g., maintaining plausible deniability). Such rapid evolution of terminology allows harmful ideologies to spread while evading detection by automated systems and, potentially, human moderators. While human moderators can effectively detect emergent hate speech terminology, they are limited in scale and can encounter vast amounts of harmful content (Jhaver et al., 2019). Generative large language models (LLMs) have been proposed as an "automoderator" for detecting hateful content. However, different LLMs disagree on what counts as hate speech and may not be able to keep up with rapidly evolving hate speech terminology, especially in the aftermath of events that may trigger a wave of hate speech on social media (Fasching & Lelkes, 2025; Garcia et al., 2025).

In this paper, we propose a context-aware computational methodology to detect emergent hate speech terminology. Though context-aware work in the hate speech sphere has previously been proposed in order to distinguish between explicit (non-coded) and implicit (coded) hate speech, the task considered in this paper is quite different from HSD, the task usually considered in the field that consists of labeling entire posts as hateful or not (Vargas et al., 2021). It is more closely related to ATE but differs from it in that such approaches are intended to discover specialized terms that align with the specialized nature of the surrounding text. In our task, the surrounding text is general, and the specialized (hateful) terms we intend to discover do not align with it. Since no previously defined task matched the one we intend to solve here, we needed to come up with a new solu-

tion. We propose leveraging our knowledge of certain ("anchor") hateful terms and using transformer-based models to contextualize their use in general language (Vaswani et al., 2017). Candidate terms are similarly transformed and their proximity to the "anchor" terms is calculated.

The process begins with a small set of known "anchor" terms associated with a specific type of hatred (e.g., the "N word" for anti-Black hatred, "kike" for antisemitism, or "tranny" for anti-LGBTQ+ sentiments). Using these anchors, the approach takes as input a large database of online posts and uses a transformer-based approach and semantic similarity measures combined with statistical estimation to identify additional terms that frequently appear in contexts similar to the anchor terms, thus indicating potentially novel or emerging hate terminology. Unlike most research in HSD and ATE, our proposed approach follows an unsupervised protocol. This was done in recognition that our task is one of discovery in a continuously evolving context, which makes the prospect of manually labeling data unrealistic. It does, however, introduce both design- and evaluation-level complexity, as will be discussed in later sections of the paper.

On the practical side, our proposed approach offers social media and policy analysts a tool for detecting the evolution and emergence of novel, emergent, or coded hate speech terminology. While the method moves beyond static keyword lists, it also preserves research agency by allowing analysts to carefully define what type of hate they are interested in analyzing through the selection of anchor terms. Thus, the approach strikes a balance between researcher-defined conceptualization and an automated method for detecting hate speech terminology.

The remainder of the paper is organized as follows. Literature Review, Part I: Hate Speech on Social Media provides background on hate speech in social media, reviews definitions, the subtle ways hate speech presents on social media, its evolution, and the general ethics of automating its discovery. Literature Review, Part II: Computational Methodology positions our research within the HSD and ATE literature, weighs the advantages and drawbacks of following an unsupervised methodology, and discusses the unique challenges this positioning poses for our testing methodology. Materials and Methods, Part I: Study Design describes the practical context in which this research was conducted, justifying the usefulness of the NHSTD task, and the design choices, including the choice to demonstrate our work on 4chan and on five different kinds of hatred. Materials and Methods, Part II: Methodology describes our app's architecture and the technical decisions behind it. Materials and Methods, Part III: Evaluation lays out our evaluation methodology in detail: how the five benchmark datasets were constructed, the baselines we pitted our app against and why we chose them, the evaluation metrics and methods used, and the division of our tests into effectiveness and expandability experiments. Results and Discussion reports and discusses our results as well as the limitations of our study. Finally, Conclusion concludes the paper and suggests directions for future work.

Literature Review, Part I: Hate Speech on Social Media

Hate Speech and Hate Speech on Social Media: Definitions

Hate speech as a phenomenon has recently seen a great deal of interest in a variety of disciplines such as law, sociology, communication, political science, and psychology (Paz et al., 2020). Paz et al. (2020) point to various proposed definitions, including "a conscious and willful public statement intended to denigrate a group of people" and "hatred toward, humiliation of, or contempt for, a 'person' who belongs to a group" (Delgado & Stefancic, 1995; European Commission against Racism and Intolerance, 2015). Some definitions add "identifying characteristics, such as race, color, religion, ethnicity or nationality", as well as "gender, sexual identity, or orientation" (Lillian, 2007; Tsesis, 2002). Ocampo et al. (2023) point to another definition adopted by the social media platform, Facebook, which states that hate speech is "a direct attack against people—rather than concepts or institutions—based on protected characteristics (PC): race, ethnicity, national origin, disability, religious affiliation, caste, sexual orientation, sex, gender identity, and severe disease" (Meta, 2022). Given that hate

speech can take many forms, Paz et al. (2020) suggest that researchers adopt a general definition that they then refine according to their particular focus.

For our work, we use an expanded definition from Meta: rather than focusing solely on direct attacks, we will also look at underhanded or indirect attacks in the form of coded terminology, where only a subset of the audience is expected to understand the utterance to be an attack. Importantly, our expanded definition retains the core requirement of hate speech: targeting based on protected characteristics. We are broadening the modes of expression, not broadening the scope to include all offensive language, which, broadly, includes vulgar or aggressive language that does not target individuals based on protected characteristics. As previous computational approaches have frequently conflated these two categories, we emphasize that our focus is specifically on hate speech (Davidson et al., 2017). The distinction is not merely definitional, but methodologically consequential. It means that the known hate terms used to guide the discovery process must be selected to reflect hatred targeting protected characteristics rather than general vulgarity, and that any evaluation of discovered terms must assess whether they carry that same targeted quality (Fortuna & Nunes, 2018; Vargas et al., 2025; Zampieri et al., 2019).

Nielsen (2002) describes different kinds of expressions of hate speech. In her paper, hate speech can be expressed at a demonstration or protest, as a symbolic gesture that does not even include speech, such as, for example, burning a cross, on websites that promote hatred or discrimination, and in exchanges between strangers in a public place. As Facebook was founded in 2004 and became available to the public only in 2006, Nielsen (2002) does not account for the modern expressions of hate speech through social media posts, YouTube videos, and so on as the article predates the era of social media.

In this paper, we specifically focus on hate speech expressed on social media in written posts. This excludes other expressions, such as hatred expressed through photos, memes, or videos, that could also appear on social media. While such expressions are extremely powerful and need to be researched as well, they are more difficult to analyze due to the technical complexity of image analysis. They also rely on extensive general and cultural knowledge to convey symbolic expressions of hatred through such media. For this reason, our study limits itself to analyzing written expressions of hatred, which can be complicated and subtle.

Hate speech on social media also embodies unique characteristics that distinguish it from hate speech more generally. The structure and algorithmic feeds of social media allow for the spread of hate speech at unprecedented speeds and volumes. The mechanisms by which platforms allow content to be quickly shared provide echo chambers for hateful content, which act as laboratories of sorts that can lead to new hate speech terminology and expressions (United Nations Office on Genocide Prevention and the Responsibility to Protect, 2023). Additionally, online hate speech can translate into tangible offline harms, such as identity-based violence where perpetrators were instigated through online hate. In the most serious cases, online hate speech can serve as an indicator of risk and a trigger for atrocity crimes, such as genocide, demonstrating how digital platforms can serve as a path from rhetoric to real-world violence against marginalized communities (United Nations Office on Genocide Prevention and the Responsibility to Protect, 2023).

Subtle Expressions of Hate Speech in Social Media

Hate speech on social media can sometimes appear under the guise of regular speech, hiding virulent sentiments using different linguistic mechanisms. This may be done for various reasons, including the intent of disseminating such sentiments to a wide audience and the attempt to evade detection and potential removal from the platform (Magu & Luo, 2018; Paz et al., 2020). In certain contexts, the use of subtle forms of hatred on online platforms is prevalent. Becker & Troschke (2023), for example, report on a study focused on antisemitism where it was found that out of 304 comments that sought to equate Israel's actions to Nazism on two news websites—Zeit Online (in German) and The Guardian (in English)—only one was expressed in explicit terms.

Subtleness or implicitness is achieved using different linguistic processes. It can occur at the word level. Magu & Luo (2018), for example, introduce the concept of Euphemistic Hate Speech whereby words are deliberately substituted for others. They cite a 2016 movement known as "Operation Google" to create code words to describe minority communities. Examples of these code words include the terms Google for Black, Yahoo for Mexican, Skype for Jew, Bing for Chinese, and Skittle for Muslim. Becker & Troschke (2023) discuss other word-level mechanisms, including the use of semiotic markers or visual signs. For example, in the word "$oros," the "S" in Soros is replaced by a dollar sign to invoke the antisemitic stereotype associating Jewish people with greed, targeting the well-known Jewish philanthropist George Soros. This transformation functions as both a visual marker and a pun, layering multiple meanings into a single typographical alteration. Another example from the same study is the made-up word "philanthrocapitalist," a play on words applied to Soros to suggest that he prioritizes profit over philanthropy.

Going beyond the word-level, Becker & Troschke (2023) argue that implicitness can occur at the sentence level using indirect speech acts such as irony or rhetorical questions. The sentence "Yes, yes, I know, Jews have never made much of money..." is an example of an ironic statement that expresses the opposite of what is being said. On the other hand, while the sentence "Do Jews always hold out their hands?" is an instance of a rhetorical question, it also uses the metaphorical phrase "hold out their hand" meant to signify "ask for money." The question is rhetorical in the sense that the speaker pretends to ask a question, whereas the speaker is making an antisemitic statement about Jewish greed. Related to this issue is the concept of dog whistle studied in Mendelsohn et al. (2023). Dog whistles use similar linguistic artifices as those described in this section (e.g., "people in our inner city" referring to Black people) but are particularly designed for political discourse. Mendelsohn et al. (2023) create a glossary of dog whistles and analyze their use in historical political discourse.

Becker & Troschke (2023) also discuss cases of implicitness that rely on shared knowledge. For example, "WWG1WGA IGWT!" can only be decoded by those familiar with QAnon conspiracy discourse, where "WWG1WGA" stands for "Where we go one, we go all" and "IGWT" stands for "In God we trust." Ocampo et al. (2023) propose a categorization of subtle and implicit hate speech based on what they term implicit properties. Beyond euphemism, visual signs, irony, metaphors, rhetorical questions, and extralinguistic knowledge, they identify 12 other properties, including inference, context, and sentiment. Regarding the inference property, Assimakopoulos & Serafis (2025) extend this categorization by proposing a logical model of the inference process using Argumentation Theory, Cognitive Pragmatics, and Relevance Theory techniques for communication.

Although subtle and implicit expressions of hate speech are quite varied, the app described in this paper focuses on word-level manifestations. Specifically, we focus on the discovery of novel coded terminology such as euphemistic substitutions, puns, and so on. This focus reflects the urgent need to address the rapid evolution of language on social media, as we discuss in the next subsection.

Regular Language and Hate Speech Evolution on Social Media

Di Marco et al. (2024) present a quantitative study of language evolution on social media over three decades. They observe that there is, over time, a decrease in both their post length and their lexical richness. Despite these trends, they also observe that individual users introduce new words to their vocabulary at a constant pace. The study does not distinguish whether these words are new to the users but already exist in the broader linguistic community, or whether they are newly created terms.

Susanto et al. (2025) address this question with a qualitative study of language evolution driven by social media. They focus on the changes in linguistic elements within various online communities and the motivations that drive these changes as well as their implications. In terms of impact, they observe that social media gives people an impetus for creativity, with the creation, for example, of new terminology as well as changes in grammar structure and language

simplification. They further note that the linguistic changes are also used to create new distinctive identities in different spaces.

Chaudhary & Asarsa (2025) illustrate some of these points with examples. In the context of new terminology, they cite abbreviations that have become words in and of themselves, such as laugh out loud (LOL), shaking my head (SMH), and fear of missing out (FOMO). Notably, these new "words" have been integrated into offline language as well. They argue that social media's speed and character restrictions have encouraged language conciseness both at the vocabulary and syntax levels. They also mention emoticons and abbreviations that not only reflect emotions, such as BRB (be right back) and TTYL (talk to you later), but also serve to build a sense of belonging within online communities. Malik et al. (2025) explore this evolution using a different methodology: they surveyed users directly about their language use when interacting with others on social media. They, too, observed the language evolution spurred by social media, as found in other studies.

All these findings pertain to the evolution of ordinary rather than hate speech on social media. However, as Becker and Troschke (2023), Magu and Luo (2018), and Ocampo et al. (2023) note, in addition to the need for conciseness and the desire to form communities as in regular social media users, there is a need for hate speech users to evade detection. This suggests an additional driver of hate speech language evolution beyond what occurs in online communities that discuss sports, music, parenting, etc. Magu & Luo (2018), Taylor et al. (2017), and Kikkisetti et al. (2024) make the same assumption and use it as a basis to build automated computational approaches to detect evolving hate speech terminology. Melillo et al. (2024) confirm that assumption by listing 24 manually discovered antisemitic terms found prior to January 2024, along with 32 novel terms discovered manually, automatically, and in a mixture of manual and automatic search on posts collected over a two-month period starting on January 1, 2024. The work was done in the context of antisemitism and instances of the discovered terms include "#FuhrerFriday," "adl sewer rats," "israhell," "talmudvision," and "satanyahu."

The rapid and ongoing evolution of hate speech terminology, combined with its sheer volume on social media, incentivizes analysts to automate aspects of detection and monitoring. However, automation in this domain raises important ethical questions, such as what counts as hate speech, who decides, and what the consequences of misclassified content are.

The Ethics of Hate Speech Detection

As Taylor et al. (2017) point out, hate speech on social media creates uncomfortable environments for users seeking civil and productive conversations. Hate speech can also damage the reputations of the social platforms on which it occurs, reducing users and revenue. Moreover, in certain jurisdictions, like in the European Union, companies are held responsible when offensive content is not removed from their platforms. However, these compelling reasons to remove hate speech on social media come with a danger: overly aggressive moderation risks suppressing legitimate discourse and threatening free speech.

Kiritchenko et al. (2021) consider the various attempts at automated HSD and observe that while results have been encouraging in experimental settings, they discriminate against certain minority groups in real-world applications. They evaluate these systems against eight different ethical principles including privacy, fairness, transparency, and promotion of human values. They consider the entire pipeline of automated hate speech detection, including dataset construction and labeling, design, implementation, testing, and deployment, and suggest these ethical considerations should be integrated at each stage.

Parker & Ruths (2023) take a different stance. They argue that while the computer science community focuses on developing increasingly sophisticated HSD techniques, key stakeholders including the government, non-profit organizations, and platforms are typically not interested in such tools due to the complex nature of the problem and the ethical and practical issues associated with them. They suggest, instead, that computer science researchers collaborate with stakeholders to learn what tools are needed and work on developing them. The approach we propose in this paper aligns with Parker & Ruths (2023). Rather than developing yet another HSD classifier, we propose a

method for identifying potential coded or non-coded hateful terminology that has emerged in hate communities. As discussed in Materials and Methods, Part I: Study Design, this problem fills a practical niche in the fight against hate speech in social media. It also addresses the ethical question of automating hate speech detection since our tool does not designate speech or speakers as hateful but rather returns a list of candidate terms for an analyst to consult and act upon.

While previous works have had different scopes—Magu & Luo (2018) and Taylor et al. (2017) targeted all categories of hateful terminology together, and Kikkisetti et al. (2024), in contrast, focused exclusively on antisemitic language—we examine five distinct types of hatred, treating each separately the way Kikkisetti et al. (2024) suggest while applying the same, unified technical framework to broaden their reach.

Literature Review, Part II: Computational Methodology

As mentioned in the introduction, within the computational methodology literature, the task we set out to solve falls neither within HSD nor ATE. Instead, it fits somewhere in-between. To clarify this positioning, we briefly review the literature in both domains. In addition, we contrast supervised and unsupervised learning methodologies and discuss the advantages and costs of settling on the unsupervised paradigm. We conclude this section with a brief overview of the evaluation strategy we developed based on our positioning.

Hate Speech Detection

Fortuna & Nunes (2018) conducted a comprehensive survey of the state of the art in automatic HSD. It carefully defines the problem, highlighting the subtle complexities of this analysis. More recently, Gandhi et al. (2024) provide a comprehensive survey of HSD methods. They note five different types of approaches (including some overlap): binary HSD, multi-label HSD, textual HSD, multilingual HSD, and multimodal HSD. Other HSD surveys describe traditional machine learning methods such as naive Bayes, decision trees, SVM models, or deep learning approaches such as recurrent neural networks, convolutional neural networks, and transformer-based models (Das et al., 2024; Kumar et al., 2022; Schmidt & Wiegand, 2017; Subramanian et al., 2023). Some of the approaches described in these surveys are very sophisticated and combine dynamic topic modeling or other contextual clustering methods to enhance the classification outcome (Choudhary et al., 2026). They also use additional contextual information, such as user profiling, and focus on more complex problems, such as multilingual or multimodal domains. Nonetheless, in the end, each of them boils down to identifying the nature of a post and categorizing it within a range of hateful or otherwise offensive bins.

Our work falls in the area of NHSTD, which aims to discover emerging terminology that is significant (though not necessarily frequent). ATE is a closer subfield, described next.

Automated Terminology Extraction

Tran et al. (2026) and Xu et al. (2025) present surveys on recent advances in Automatic Term Extraction. Tran et al. (2026) note that "Automatic term extraction (ATE) is a Natural Language Processing (NLP) task that eases the effort of manually identifying terms from domain-specific corpora by providing a list of candidate terms (p. 1)." They further state that "Terms are textual expressions that denote concepts in a specific field of expertise (p. 1)." Xu et al. (2025) describe traditional ATE approaches as linguistic-based, statistics-based, and hybrid. More recently, they note, the field has moved toward newer paradigms, including machine learning, deep learning, graph-based, and language model approaches (which encompass transformer-based and large language models). Tran et al. (2026) make clear that ATE differs from NHSTD, but several of its approaches can be adapted. Specifically, ATE typically follows a two-step pipeline: identify candidate terms, then classify them via supervised or unsupervised learning approaches. NHSTD can adopt this pipeline. The approach we propose draws on statistical and contextual embedding methods covered in these surveys.

Two methods stand out as adaptable to NHSTD: Bennani-Smires et al. (2018) and Sharma & Li (2019). Bennani-Smires et al. (2018) use part-of-speech tagging to extract candidate terms and then compare

these terms' Sent2Vec/Doc2Vec embeddings to document embeddings of their database using cosine similarity. Sharma & Li (2019) improved this pipeline with BERT, embedding both N-grams and documents. Variations on this theme led to KeyBERT (Grootendorst, 2020), now a leading ATE approach, especially for short texts (Nadim et al., 2023).

These approaches are inspiring, but do not fully fit our task. We are not trying to discover hateful terminology that resembles a post or set of posts. Instead, we use extrinsic information—terms already known to represent a particular type of hatred, along with the posts in which they appear—to form their embeddings. The closest prior work is King et al. (2017), the only other study we found that explicitly leverages extrinsic knowledge, although the paper's methodology does not use modern NLP techniques. Rather, they use known event- or position-related terms to retrieve relevant documents, from which they then discover keyphrases. Our approach uses the extrinsic knowledge directly.

Beyond our direct injection of extrinsic knowledge, our approach also improves efficiency. Specifically, we adopt a hybrid approach: Term Frequency-Inverse Document Frequency (TF-IDF) first filters candidate terms, then contextual embeddings and semantic comparison are applied to the remaining terms. Prior approaches instead rely on part-of-speech tagging, generate all possible N-grams, or search all possible keywords without differentiating on relevancy, which applies expensive embedding computations to far more candidates than necessary.

Finally, we chose an unsupervised approach to bypass the bottleneck inherent to supervised methods. We discuss this tradeoff in the next subsection.

Supervised versus Unsupervised Learning

Supervised learning relies on labeled data, whereas unsupervised learning uncovers structure without prior labels. This makes unsupervised methods more flexible for discovering new knowledge, whereas supervised learning largely reproduces and generalizes what is already labeled. That said, when the goal of the learning process is clear and the desired categories are well-defined, supervised learning typically outperforms unsupervised approaches (Bellinger et al., 2017).

Almuqati et al. (2024) review the challenges of supervised and unsupervised learning. Two are particularly relevant here: the need for labeled data (supervised) and the difficulty of evaluating results (unsupervised). Although we previously argued that our task of NHSTD is inherently different from the task of HSD, one could conceivably treat it as supervised: extract the context of each candidate term from Step 1 and classify the term as hateful or not based on that context, as Hoeken et al. (2024) propose. In theory, this is feasible—a labeled dataset of hateful terms in context could train such a classifier. In practice, it is not. Hoeken et al. (2024) labeled 4,000 terms in context, and hate speech data is notoriously difficult to design and label, especially on social media, where context and terminology evolve rapidly (Zampieri et al., 2019). A supervised approach would therefore require periodic retraining on freshly labeled data, sacrificing the flexibility and adaptability that unsupervised methods provide.

The same flexibility that lets unsupervised learning discover new knowledge also makes it difficult to evaluate comprehensively. Supervised learning splits labeled data into training and test sets and scores predictions against a gold standard; unsupervised learning has no such gold standard, which leaves discovered results difficult to assess. We therefore designed the evaluation strategy sketched out next and developed more thoroughly in Materials and Methods, Part III: Evaluation.

Overview of our Evaluation Strategy

The first issue—unique to NHSTD and not encountered in HSD or ATE—is that NHSTD is not an established task. We defined it for the hate speech analysis community based on a needs assessment. Hate Mitigation App (HMApp) is the only app of its kind, making direct comparisons with state-of-the-art (SOTA) approaches impossible. To show that NHSTD genuinely cannot be handled by existing methods, we selected one of the strongest methods from ATE, the closest family to NHSTD, as our SOTA baseline.

The novelty of the NHSTD task also creates a second issue absent from HSD and ATE research: the lack of benchmark datasets. Both NHSTD and ATE are discovery tasks, but ATE has existed long enough—the first comprehensive survey appeared in 1996 (Kageura & Umino, 1996)—that benchmarks now exist to simulate discovery. NHSTD has none. We therefore created five: the first validates our approach thoroughly, and the others test its expandability.

A third issue is that NHSTD, unlike HSD, is not a classification problem. Classification systems aim to label every class well; NHSTD, like ATE, aims only to discover the terminology of one class, with the others irrelevant. As Šajatović et al. (2019) note in their discussion of ATE evaluation, ATE and NHSTD systems output ranked lists of candidate terms with a "goodness" score rather than class assignments. We addressed this in two ways: by choosing cutoff points and by examining performance across all possible cutoffs.

Materials and Methods, Part I: Study Design

This section aims to contextualize our approach. We seek to develop a tool that enables hate detection and mitigation researchers and practitioners to quickly discover emerging or trending hateful terms, including both coded and non-coded language, as well as to detect terms that may have been hidden for a while. Rather than offensive terms of a general nature, our focus is on hate speech that targets specific groups. The tool is designed to be user-friendly and versatile, capable of both analyzing user-gathered datasets and standardized data collected by our system. While practitioners in this area may recognize certain hate terms, they are often unaware of emerging or coded expressions that evolve rapidly, making such a tool essential.

Rather than searching for hate terms indiscriminately, we chose to treat each category of hatred separately, analyzing them one at a time so that emerging terms are identified in the specific context of each category. Our results indicate that these categories frequently overlap, as illustrated by this post from 4chan retrieved by our app when searching for anti-Asian hatred (warning: the post contains strong, hateful language):

> I'm white, and yes, I am trying to spread more hate. TOTAL JEET, NIGGER, SPIC, GOOK, CHINK, KIKE, SANDNIGGER DEATH

In this post, each capitalized word between "TOTAL" and "DEATH" is a derogatory term for Indian, Black, Hispanic, East/Southeast Asian, Chinese, Jewish, and Arab/Middle Eastern/Muslim people, respectively.

In addition to surfacing trending terms, the app can display the context in which they occur, offering insight into the psychology or sociology behind hateful utterances. Consider the following example, also from 4chan (warning: the post contains strong, hateful language):

> Canada has been in decline for fucking 40 years but the government won't admit it. Despite importing every chink, flip flop and pajeet top 0.1% engineer our GDP per capita has been trending down.

Tracking the evolution of hate speech on social media is valuable across multiple sectors. In law enforcement and crime prevention, hate speech on social media often precedes physical actions and violence (Binder & Kenyon, 2022; Kalfas, 2025; Laub, 2019; Williams, 2019). Mental health services can use this understanding to detect distress in individuals (Walther, 2022), deploy appropriate interventions, and reach out to victims of online hate speech (Saha et al., 2019; Shmulewitz et al., 2025). In the social and political sciences, knowledge and understanding of coded discourse is also useful to assess a range of questions such as the social effects of coded language, the practice of code switching to dispel negative perceptions in certain contexts (Johnson et al., 2022), and the role of coded language (or dog whistles in this case) in democratic states (Blackington & Cayton, 2025; Tillyris, 2024). Finally, for both automated (Horta Ribeiro et al., 2023; Kolla et al., 2024) and human-machine collaborative content moderation (Jhaver et al., 2019), maintaining current knowledge of hate speech terminology is essential for effective platform governance. Indeed, practitioners rely on these terms as an initial filter that can be thought of as a first line of defense and, when augmented with explanations, they can provide human-understandable justifications (Polignano et al., 2022).

Using 4chan Data

The HMApp's default data source is 4chan's /pol/ (Politically Incorrect) board, an anonymous image-board with minimal moderation that has become a widely studied fringe platform in the hate speech literature. This choice is motivated by two complementary observations: /pol/ harbors an unusually high concentration of hate speech, and the hateful language that originates there demonstrably migrates to mainstream platforms.

In terms of prevalence, Hine et al. (2017) conducted a large-scale measurement study of /pol/, collecting over 8 million posts and finding that hate speech and extremist content were widely found on the board. Rieger et al. (2021) found that 24% of sampled /pol/ comments contained at least one form of hate speech, compared to 13.8% on the more mainstream subreddit /r/The_Donald. More recently, Bermudez-Villalva et al. (2025) used transformer-based models to classify /pol/ posts and reported that 11.2% contained hate speech spanning racism, religious hatred, sexism, and anti-LGBTQ+ content, with racism alone accounting for 35.9% of hateful posts.

Crucially, the hateful content produced on /pol/ does not remain confined to 4chan. Hine et al. (2017) documented coordinated "raids" in which /pol/ users targeted YouTube videos, producing a measurable increase in hate speech within the comment sections of those videos. Zannettou et al. (2017) analyzed millions of posts across Twitter, Reddit, and 4chan and found that alt-right communities on 4chan and Reddit exerted a strong influence on Twitter. They provide evidence that fringe communities regularly spread alternative content to mainstream social networks. A follow-up study by Zannettou et al. (2018) traced the propagation of 160 million images across the same platforms, again finding that /pol/ and similar fringe communities function as originators of memes that subsequently diffuse to mainstream platforms and audiences.

Taken together, these studies demonstrate /pol/ as a laboratory for the creation of novel hate speech terminology and a vector through which such terminology reaches the wider social media landscape. This dual role makes /pol/ an ideal data source for an app designed to detect emerging or previously hidden hateful terminology before it becomes widespread. By monitoring /pol/, the HMApp can surface coded and emerging terms at or near the point of origin, providing researchers and practitioners with an early warning capability that would be difficult to achieve by monitoring only mainstream platforms where such terms appear later, often in a diluted or remixed form.

The Five Categories of Hate

The HMApp currently supports five hate categories: antisemitism, anti-Black hatred, Islamophobia, anti-Asian hatred, and anti-LGBTQ+ hatred. These were selected based on two considerations: their empirical prevalence in our data source, and the urgency conferred by recent events.

First, these categories align with the empirically observed dominant forms of hate speech on 4chan's /pol/ board. Bermudez-Villalva et al. (2025) found that racism constituted 35.9% of hateful posts, religious hatred, directed primarily at Jewish and Muslim communities, accounted for 23.3%, anti-LGBTQ+ content comprised 16.5%, and sexism represented 12%. Our five categories thus map directly onto the most prevalent dimensions of hate on the platform from which the HMApp draws its data.

Second, each category has been affected by recent events that could generate and spread novel terminology, which is precisely the kind of language the HMApp is designed to detect. The October 7 Hamas attack and the subsequent conflict in Gaza triggered sharp spikes in both antisemitic and Islamophobic online discourse (Rosen & Walther, 2025). The COVID-19 pandemic gave rise to anti-Asian slurs tied to the virus's geographic origins (He et al., 2022; Hswen et al., 2021). Legislative debates over LGBTQ+ rights, particularly concerning transgender individuals, have generated new coded vocabulary (GLAAD, 2024). The Black Lives Matter movement and the murder of George Floyd in 2020 triggered a 250% spike in online racist posts (Lupu et al., 2023), and anti-Black hatred more broadly continues to produce new euphemistic and coded terminology in response to social and political events. These event-driven surges make the selected categories especially productive sites for studying novel terminology in real time.

We acknowledge that these five categories do not cover the full spectrum of identity-based hatred. Forms such as misogyny, anti-disability bias, and xenophobia directed at immigrant communities, especially Hispanic and Latino populations, are all significant and warrant attention. The HMApp's architecture is designed to be expandable: adding a new category requires only the specification of appropriate anchor terms and does not necessitate changes to the underlying technical framework (see "The Technical System Architecture"). We view the current five categories as a principled starting point that balances breadth of coverage with depth of evaluation, and we plan to extend the system to additional categories in future work.

Having established the context for our work, we now turn to the methodology used to design and implement the app, its evaluation, and planned future developments.

Materials and Methods, Part II: Methodology

In this section, we describe the HMApp.

The Technical System Architecture

Figure 1 displays the HMApp architecture, which uses the following color scheme. User-entered information is shown in orange, while the embedding vectors of the anchor terms corresponding to the hatred selected by the user and discussed in Vector Database for Semantic Analysis are shown in blue. The intermediate results produced by the pipeline's internal components are displayed in light marigold, while the app's output is shown in the pink box. Arrows illustrate the movement from one stage of the pipeline to another. They are labeled with the name of the procedure used to make this move possible. We now explain the details of the pipeline starting from left to right on the upper side of the figure.

The user begins by feeding the app seed words already known to represent hatred of a particular category. A keyword search is then performed using these keywords. This results in a set of posts that contain the seed words entered by the user. A procedure called TF-IDF (see "Identifying Trending Terms") is then applied on the set of posts to retrieve terms considered "important" from a relative frequency point of view. It is worth noting that the semantics of these terms have not yet come into play at this stage.

To include semantic considerations, the most relevant candidate terms are turned into numerical vectors (or embeddings) that do take semantics into account through the BERT transformer (see "Vector Database for Semantic Analysis"). This results in a set of embedded candidates that are organized in a way that brings words with similar meanings (because of the similarity of their context) together and separates those with different meanings (based on their difference in context). None of this organization, however, is interpretable by human beings.

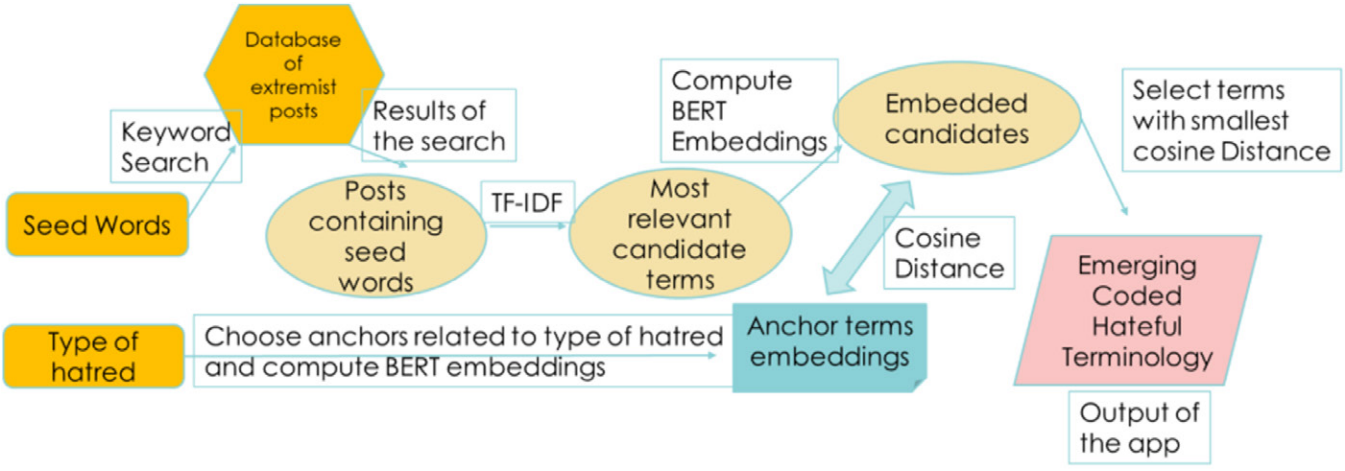


Figure 1. The HMApp pipeline

To make the organization interpretable, the numerical vectors need to be pegged to numerical vectors for which there is a current human understanding. The bottom of the graph displays the process where the pegging of numerical vectors to terminology is done. For each type of hatred considered, we have, in advance, researched the terminology that is already known to express that hatred, picked a set of representative terms, and embedded them in a Vector Database (see “Vector Database for Semantic Analysis” and “Candidate BERT Embeddings and Hate Relevance Score”).

Having described all the preliminary pieces of the pipeline, we can now explain how the app proceeds to extract emergent hateful terminology. The process is represented by the blue double arrow linking the bottom part of the graph to the top. As previously mentioned, the embedding process central to the app created a mathematical space within which words with similar contexts are grouped together while those with different contexts are kept apart. By measuring the distance of the embedded candidates found by the procedure on top of the graph to the anchor terms embeddings at the bottom of the graph, we can decide which candidate terms are likely to represent hatred and which are not. Indeed, the closer the candidate terms are to the anchor terms, the more likely they are to be hateful themselves. This proximity measure (the cosine distance) is used to select emerging hateful terminology.

Navigating Through the HMApp User Interface

We briefly describe how users can intera ct with HMApp. Figure 2 displays the HMApp user interface input panel. The panel illustrates the different choices that a user can make. First, the user can choose a data source mode. Two options are available: users can choose their own database (in CSV or Excel format) with at least one column, the text column, or they can use the HMApp’s default database. As previously described, the default database is the 4chan / pol/ feed; our approach is flexible enough that additional social media sources can be added in the future. The database is updated daily using a 4chan feed by running internal Python scripts. In Figure 2, the user has selected the default option. The second choice concerns the selection of a type of hatred. We support five hate categories for the time being: antisemitism, anti-Black hatred, Islamophobia, anti-Asian hatred, and anti-LGBTQ+ hatred. Third, the user is invited to enter the seed terms around which the search will revolve, along with the date range meant to restrict the search to posts falling within that time period. Though

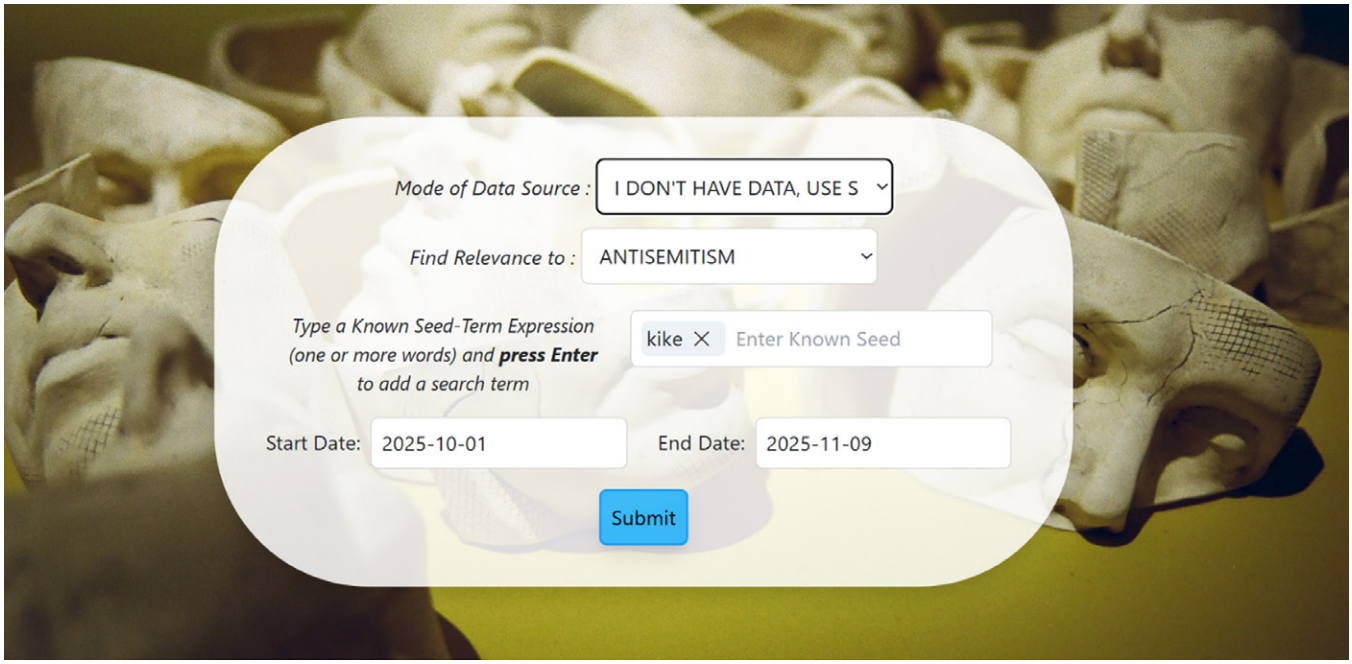


Figure 2. The HMApp user interface

the user can enter any seed term, the process will not be successful if the seed terms are not related to the type of hatred previously selected.

Once the user presses the "Submit" button, the methodology described in "The Technical System Architecture" is applied to the database to produce the emergent hateful terminology output. The output terminology is incorporated into one of the panels on the screen, as shown in Figure 3. Initially, five terms are presented on the screen, sorted by their relevance to the specified type of hatred and accompanied by relevance and frequency scores. The user can view additional terms by clicking the "Get 5 More Terms" button to load the next set and can continue pressing the button until all terms are displayed. The seed terms are not included in the expressions listed on the screen, but the user has the option to apply a filter to view them or expressions containing them as well.

Another filter lets users sort terms by frequency. The user can also click on terms or expressions of interest to view the other two panels on the screen. One panel, on the right side of the terms list, displays the evolution plot of the selected terms over the initially chosen date range. Figure 3 shows the evolution plot for the term "psyop" (a term that stands for a military "Psychological Operation" to influence the thoughts and emotions of the target audience; it is often used in these posts to suggest that Jews engage in such operations, related to well-known antisemitic tropes such as greed and control) that was selected from the list of words on the left. The user can also request to view the plot for the 30 days preceding the start date and the 30 days following the end date to see how the terms evolved during those periods as well. The second panel at the bottom of the screen, shown in Figure 4, contains the actual posts that use the selected terminology. A researcher engaged in antisemitism mitigation could use the contexts of this term to better understand newer variations of these tropes.

Additional user interface features include the ability to download the plot, export the Excel file containing the chosen terms and their related posts, create a new session with the selected terms as input for the next run, and restart the session. We are also in the process of adding more functionality, such as AI explainability features and an authorization and authentication method.

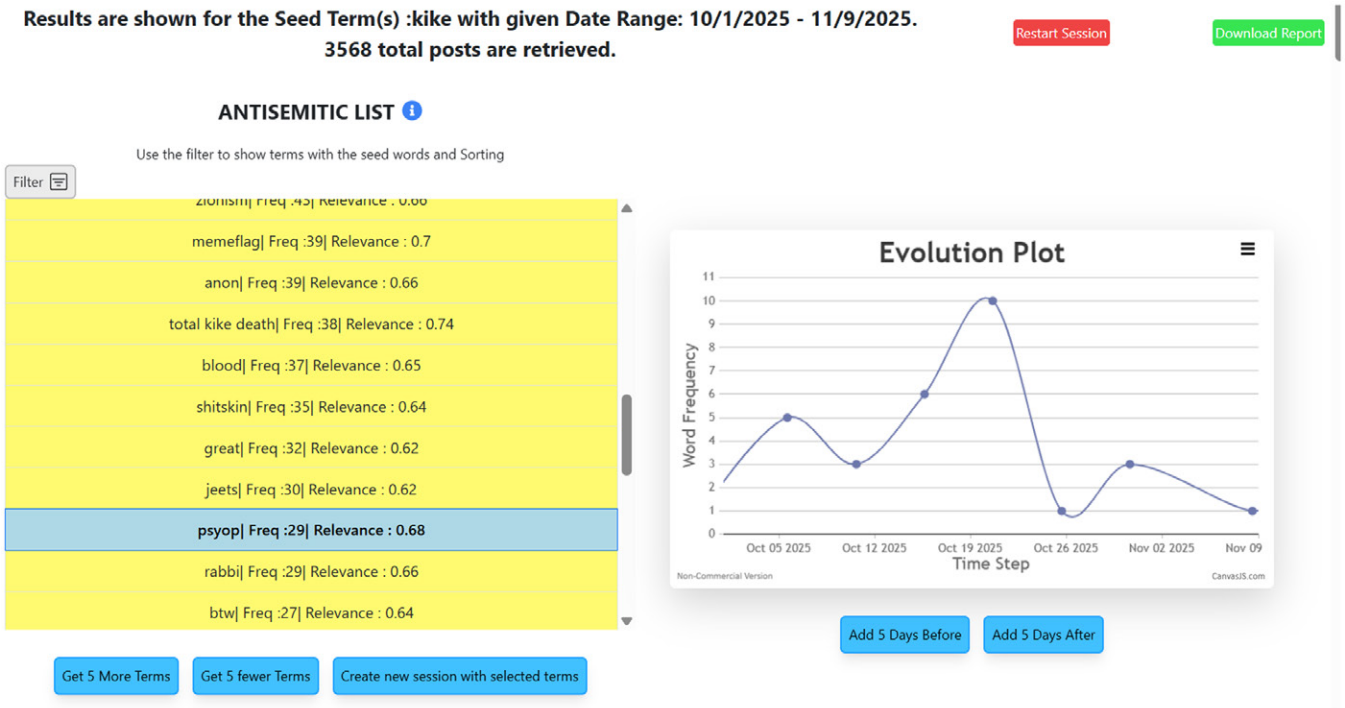


Figure 3. The HMApp's first output screen

We now turn to more detailed and technical explanations of how the pipeline of Figure 1 was implemented.

Technical Specifications

This section explains the technical details needed to discover potential emerging novel hate terms from posts containing a user's seed words. We begin with a description of how the posts are initially cleaned and conclude with the formula that calculates the relevance score of a term to a specific kind of hatred.

Identifying Trending Terms

As mentioned in the previous section, the user selects a data source, a type of hatred, seed words that embody or represent that hatred, and a date range. This produces a subsetting of the original database tailored to the user's choices. Within that smaller database, the data posts are raw, uncleaned, and contain noise that must be pre-processed before any analyses. Several standard data preprocessing steps are first applied, including the removal of hyperlinks and special symbols, the conversion to lowercase, and the removal of stop words (very common words that occur in virtually all posts, such as "the" and "a"), among other procedures that make the data more suitable for NLP tasks. Unigrams (single words), bigrams (two-word expressions), and trigrams (three-word expressions) are formed by applying one-, two-, and three-word windows to all the pre-processed posts. Badly formed expressions are then filtered out using a part-of-speech tagger to identify the grammatical function of each word in the expression and remove those that do not conform to pre-determined acceptable grammatical forms. Specifically, expressions containing verbs, adverbs, prepositions, and certain other word categories are eliminated.

TF-IDF scores are then calculated for each term or expression in the given posts. TF-IDF is used to measure the importance of terms or expressions within a document and across the collection of documents. In related work, Kikkisetti et al. (2024) find that TF-IDF outperforms concordance and collocation when used in a similar pipeline.

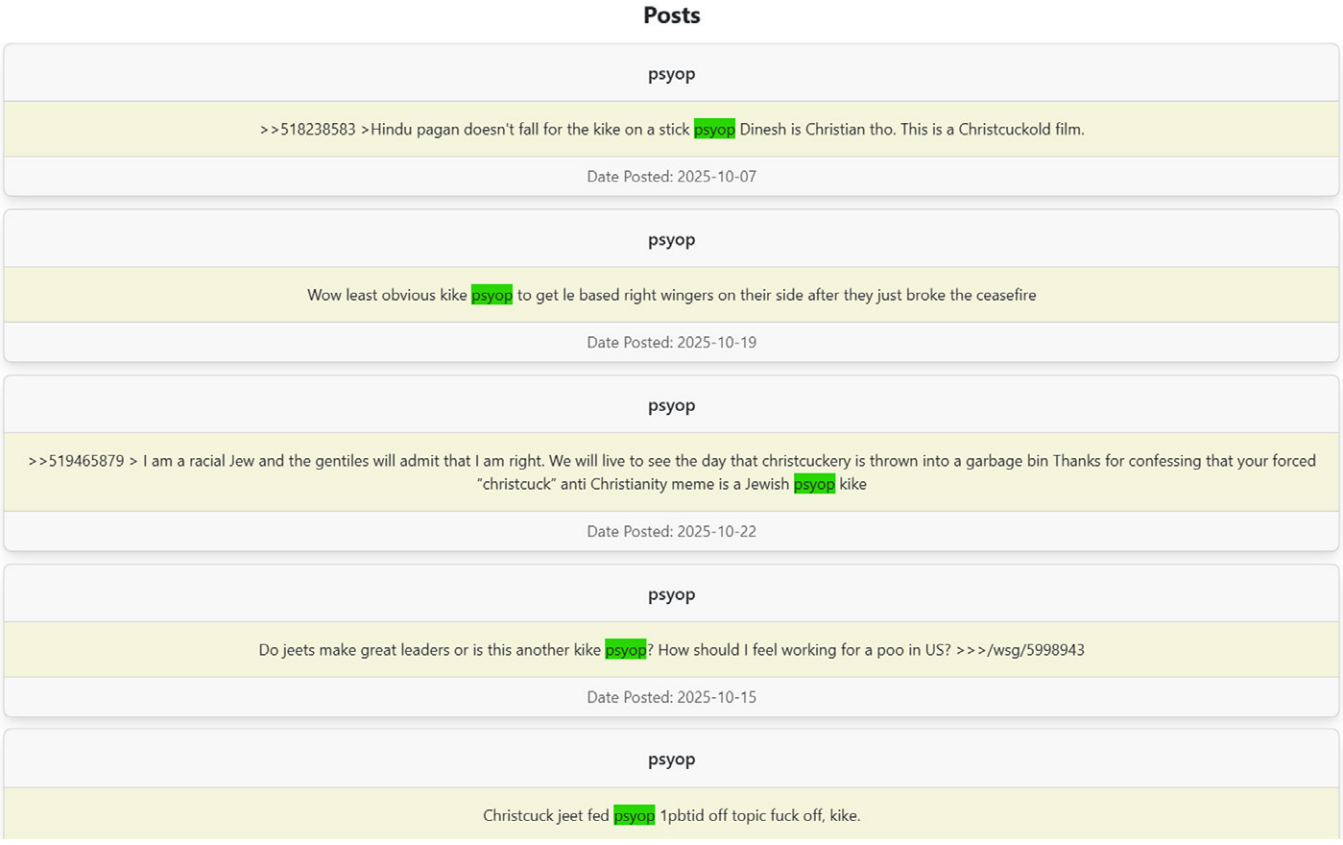


Figure 4. The HMApp's second output screen

Candidates are retained if the individual term's TF-IDF score exceeds an analyst-defined threshold. We use the same threshold defined in Kikkisetti et al. (2024): for each term, we take its highest score across all documents; the threshold is set to the average of these per-term maximum scores. This is a frequency-based method, and, as previously mentioned, no semantic notions are involved until candidate terms are filtered. All important candidate terms—those with high TF-IDF scores—are considered for the next phase of identifying potential emerging hate terminology.

Vector Database for Semantic Analysis

We now discuss the procedure for creating lists of highly representative anchor terms for each type of hatred (as described at the bottom of Figure 1). The same general procedure is followed to curate the known hate terms for each type of hatred, though slight changes were needed for specific types or terms. We begin by discussing the full procedure, then describe which part of it was actually applied in each of the specific cases we considered.

General Procedure. The procedure involved first selecting terms using authoritative sources. When the reliability of the initial sources was uncertain, or the preliminary procedure retained too many slurs, the terms were cross-referenced against two well-known hate speech data sources, Hatebase and the Racial Slur Database, and only terms present in at least one were retained.[1] For Hatebase, only terms labeled "Highly Offensive" or "Extremely Offensive" were retained. Terms absent from Hatebase were checked against the Racial Slur Database when possible, and any matches were moved forward.

If the list was still long, the next step consisted of gathering each potential anchor term's frequency of usage on social media over the past 6 months. This was done using Pyrra Technologies (now part of AlertMedia), a service that aggregates and analyzes content from dozens of social media platforms.[2] However, this count could not be taken directly: some terms can appear in general contexts that do not convey hatred as well as in coded forms that do, and tallying both would weaken our methodology. To prevent this, we used Pyrra's feature that isolates posts classified as Offensive, Violent, or Hateful, and gathered frequency data exclusively from those filtered posts. The terms were then sorted in descending order of frequency, and a cumulative-proportion threshold was applied to select the most representative ones.

Application to Specific Kinds of Hatred. We selected anchor terms using specialized resources for each type of hatred. The full list of terms is in the Appendix.

For anti-Black hatred, we drew an initial list of 89 terms from Wikipedia's list of slurs (filtered to English-language terms targeting the Black community), narrowed it to 39 via Hatebase and the Racial Slur Database, and then to 22 final anchors using the Pyrra frequency step with a 95% cumulative-proportion threshold, i.e., the final anchor terms are retained if they contribute to the top 95% of the cumulative post frequency across all terms when the terms are sorted in descending order of overall frequency.[3] For antisemitism, we used the 20 explicitly antisemitic terms and tropes from the American Jewish Committee's Translate Hate Glossary directly, without the Pyrra step, because the American Jewish Committee (AJC) explicitly flags these terms as always antisemitic and highly offensive.[4] For anti-Asian hatred, we combined the Committee of 100 glossary with anti-Asian slurs from Wikipedia, yielding 45 initial terms that the Hatebase and Pyrra checks reduced to 16.[5] For anti-LGBTQ+ hatred, we drew roughly 30 slurs from the Chew Glossary curated at the National Library of Scotland, retaining 24 after the frequency

1 The Racial Slur Database was active when we created the anchor terms for anti-Black hatred and antisemitism, but was unavailable when working on the other three types of hate. As a result, we only cross-checked the anti-Asian, anti-LGBTQ+, and Islamophobic terms against Hatebase.

2 Pyrra Technologies was recently acquired by AlertMedia: https://www.alertmedia.com/products/social-media-intelligence/.

3 https://en.wikipedia.org/wiki/List_of_ethnic_slurs

4 https://www.ajc.org/translatehate

5 https://www.committee100.org/our-work/anti-hate-glossary/

analysis.[6] For Islamophobia, no comparable institutional resource was available. We first used Wikipedia to obtain 16 terms, but only one passed the Hatebase evaluation. We used five additional terms from Mustafa et al. (2025) directly, leaving us with an anchor set of six terms.

Constructing the Vector Databases. Once all the anchor terms were selected, vector databases were created for each kind of hatred. A vector database stores the context of each anchor term's usage in the posts. This is done by building the contextual embedding of each anchor term using the procedure described in the next section for the contextual embedding of candidate terms: the procedure is the same whether a term is an anchor term or a candidate term, except that for anchor terms, the posts used are those retrieved, in advance, from Pyrra to compute the frequency of each of the anchor terms instead of the posts retrieved by the app.

As previously shown in the pipeline, the context of terms extracted by the TF-IDF procedure is subsequently compared with the context of these anchor terms to identify potential hate terms. A different vector database is created for each kind of hatred. ChromaDB, an open-source vector store, is used for storing anchor term embeddings and conducting similarity searches. It employs Hierarchical Navigable Small World graphs, a prevalent algorithm for approximate nearest neighbor searches, to facilitate semantic search processes.

Candidate BERT Embeddings and Hate Relevance Score

We now describe how BERT embeddings are used and how the relevance (or proximity) score is computed. As previously alluded to, the BERT transformer is one of the neural network-based approaches that projects words and their contexts into a numerical space in which words with similar meanings are grouped closer together than those with distinct meanings. This technique allows synonyms to be placed closer together and homonyms farther apart. For example, this technique organizes words to show that the "river" meaning of the word "bank" is related to the word "waterfront" whereas the "financial" meaning of the word "bank" is related to the word "credit union" and the two meanings are located far away from one another in the numerical space calculated by BERT. This is a useful characteristic when dealing with coded terms as well. For example, "ZOG" is a coded term that, on the one hand, stands for "Zionist Occupation Government," a highly offensive antisemitic expression, whereas it is also the name of a friendly dragon in children's books. Using a BERT embedding approach allows us to identify two "ZOG"s in different areas of the space, thus distinguishing between the offensive and non-offensive meaning of the term.

We describe the steps we followed to calculate the relevance of a single potential term for one of the hate vector databases, H (where H represents anti-Black hatred, antisemitism, or any of the other three kinds of hatred studied here). First, we generate the contextual embeddings within a post for each occurrence of the potential hate term $t \in T$, where T represents the set of all potential hate terms. By contextual embedding, we mean the embedding specific to the term's usage, not the embedding of the entire post. In other words, we only consider the words surrounding the term rather than the entire post. If the term appears multiple times within the same post, we compute the average of those embeddings to obtain a single embedding for that post. Averaging ensures that each post yields an embedding of the term of the same dimensionality, regardless of how many times the term appears. If we concatenated these embeddings, we would get embeddings with different dimensions. If the term appears in n_t different posts, then n_t separate embeddings are generated. Currently, a general BERT model is used to calculate these embeddings because Kikkisetti et al. (2024) show that BERT outperforms specialized models such as HateBERT (Caselli et al., 2021). However, this may change depending on the results of testing with other models to determine which provides the most effective results.

To compute proximity to hateful anchor terms from hate database H and calculate relevance scores to that hatred, we compare p_{it}, the embedding of term

6 https://itg.nls.uk/wiki/Introduction

t in the ith post, to all the entries of the vector database H, and select the top S relevance scores, where relevance is defined as cosine similarity. In the current iteration of our application, S=5. This choice balances two limitations. Setting S=1 ties the relevance score to a single nearest anchor, making it sensitive to spurious matches; averaging over all anchors dilutes the signal, since a coded term targeting a specific stereotype may legitimately only resemble a subset of the anchors. Averaging the top S=5 matches smooths over the noise from any single anchor while concentrating the score on the most contextually aligned ones. That said, the optimal value of S remains an open question that we leave to future work, particularly for categories like Islamophobia, where the small anchor set may warrant a category-specific choice.

We then compute the average of these S cosine similarity scores to obtain a single relevance score r_{it} for the term t in the ith post. This process is repeated for all embeddings to obtain n_t individual relevance scores. We then calculate the overall relevance score for the potential term t as:

$$R_t = \frac{\sum_{i=1}^{n_t} r_{it}}{n_t}$$

The procedure is applied to all potential terms, resulting in a set of |T| relevance scores. Terms with scores greater than the average of all relevance scores are selected and returned to the front end; that is:

$$\left\{ t \in T : R_t > \frac{1}{|T|} \sum_{i=1}^{|T|} R_i \right\}$$

Having described the functionality and some of the technical details of the app, we now turn to its evaluation.

Materials and Methods, Part III: Evaluation

An app's design and implementation cannot be considered complete until it passes tests that demonstrate that it does what it was built to do and does so better than existing methods. To evaluate our app, we designed a methodology that tests, benchmarks, and validates our generalized hate speech terminology identification application. The process integrates authoritative hate speech resources, contemporary usage data, and comparisons with state-of-the-art automated terminology extraction approaches as well as LLMs to ensure comprehensive evaluation.

As noted earlier, three factors complicate this evaluation: the absence of benchmark datasets, the absence of directly comparable software, and the fact that our app is built for discovery rather than classification.

This section expands on the methodology briefly discussed in "Overview of our Evaluation Strategy". We begin with the construction of our benchmark datasets, then describe TF-IDF and KeyBERT, the methods we selected for comparison. We follow with the evaluation metrics and reporting methods, and conclude with the experiments we conducted to assess the HMApp's effectiveness and expandability.

Construction of Our Benchmark Datasets

We constructed five benchmark datasets corresponding to five types of hatred: antisemitism, anti-Black hatred, Islamophobia, anti-Asian hatred, and anti-LGBTQ+ hatred. Because our approach is unsupervised, these datasets are used solely for testing. Below, we describe the distinction between Hard and Soft scoring followed by a detailed account of how the datasets were constructed.

Hard and Soft Scores

For each dataset, we devised two ways of scoring the results. We refer to these as "scores" with the understanding that specific metrics, discussed later, serve as the actual scoring methods.

The first one, the "hard scores," compares the terms returned by the app against established glossaries of hatred. These metrics are the least susceptible to bias because the glossary terms were typically compiled by researchers and professionals who reached consensus or had justification for each term's inclusion. The trade-off is that this approach only confirms the app returns terms already recognized as hateful; it says nothing about its practical usefulness, its ability to discover new terminology. Users concerned only with known terms could simply consult the existing lexicons.

This motivates the second evaluation, the "soft score." Here, the app's output is verified against more flexible, adaptive sources identified through manual web searches for current information. This score is inherently susceptible to bias because our searches may not be exhaustive, and we are relying on web content that lacks authoritative certification of the resources used in the hard score. However, it incorporates current information, providing a measure of accuracy not provided by the hard score. The soft score is therefore a measure of the app's practical utility: while the terms may exist elsewhere on the web, users would be unlikely to find them without the app to surface them (see the Appendix for results of searches for emergent antisemitic terms on Perplexity AI and Google Gemini).

The third evaluation method is qualitative and, consequently, more susceptible to bias. Our assessment is that the soft score still does not convey the full effectiveness of the app. For that reason, we listed terms we judged to be offensive toward the minority group in question, but that neither the hard nor the soft score could evaluate. This assessment relied on our domain expertise in hate speech research to determine whether the extracted terms constituted hate speech.

Benchmark Data Set Construction

The construction of our benchmark datasets was done in the following three steps.

Step 1: Selection of Reliable Reference Sources. To calculate both hard and soft hate-association scores, we curated a set of authoritative, contemporary, and domain-relevant resources. For general hate speech, we used Hatebase, the non-latent semantic lexicon, and HurtLex. For antisemitic content, we supplemented these with the AJC and the Anti-Defamation League's Antisemitism Perception Tool.

Hard score calculations rely exclusively on these verified sources to ensure definitional accuracy. Soft score calculations, in contrast, draw from public online discourse—blogs, Reddit discussions, and news media—to capture contextual or emergent associations with hateful language.

Step 2: Selection of Seed Terms for Collecting Posts. To populate the system with posts containing potentially novel hate terms, we collected seed terms from relevant Wikipedia entries. Each potential seed term was checked against Pyrra's frequency data for the preceding 30 days. The ten most frequently occurring terms were selected.

For each seed term, our approach retrieved up to fifteen of the most relevant potential racial or hateful terms. We used a primary date range of the posts from October 9 to October 26, 2025. If fewer than fifteen terms were returned, we expanded the date range of the posts from May 9 to October 26, 2025. In rare cases where fewer than five terms were retrieved, we used the full range of the posts from December 19, 2024 to October 26, 2025.

Step 3: Extraction and Scoring of Candidate Terms. From the generated lists, the top fifteen terms for each seed were selected for scoring. For our hard score, each term was queried across all relevant authoritative sources as previously described. We assigned a score of 1 if a match was found (including partial matches for multi-word expressions), and a score of 0 otherwise. For the soft score, each term was queried using Perplexity AI with the following prompt: "Is this word related to [type of hate] discourse online? both coded and non coded: [the word]." We assigned a score of 1 if any referenced source identified the term as hateful, and a 0 if none did.

This three-step process was repeated for all five different kinds of hatred considered in this work, yielding five different benchmarks.

Selection of State-of-the-Art Methods for Comparison

To assess HMApp's effectiveness, we compared it against closely related methods. As discussed previously, NHSTD is a new task with no obvious competitor. However, we identified two appropriate ATE approaches: KeyBERT (Grootendorst, 2020), a neural-based approach, and TF-IDF, a keyword-based approach (Salton & McGill, 1986).

KeyBERT is a popular and effective keyword extraction method. Like HMApp, KeyBERT is unsupervised and uses BERT embeddings and cosine similarity; it is also considered an overall leader in keyphrase extraction, especially on short texts (Nadim et al., 2023). KeyBERT identifies key terminology by embedding the entire

sentence as a reference. Because the posts are associated with a specific type of hate, the extracted keywords are expected to reflect that context, even though, unlike our approach, KeyBERT does not use anchor terms.

TF-IDF is an older method that is purely statistical and does not use any of the modern neural technology. It is designed to evaluate how important a word is to a particular document within a larger collection of documents. In our application, the larger collection of documents represents all available posts, while a particular document corresponds to a single post of interest. Unlike HMApp and KeyBERT, TF-IDF uses no notion of context or semantics. In fact, TF-IDF is not foreign to the HMApp, as it is used in its first step, as discussed in Materials and Methods, Part II: Methodology.

Choice of Evaluation Metrics and Methods

We now discuss the complex task of selecting appropriate evaluation metrics and methods, along with the choice of cutoffs.

Evaluation Metrics

In a typical information retrieval setting, the three main metrics of interest are precision, recall, and their combination, the F-Measure (specifically, we use the F1 score). They are defined with respect to the entries of a confusion matrix: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN). The metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$\text{Recall} = \frac{TP}{TP+FN}$$

$$F_1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

While these metrics are straightforward to calculate in classification tasks, where every instance is assigned to a class, they are less so in the ATE or NHSTD context. These methods return only the terms they consider members of the "positive" class, ignoring any terms that are part of the "negative" class. This makes TP and FP—and thus precision—calculable, but leaves TN and FN—and thus recall—out of reach. To recover recall for a full effectiveness evaluation, we need to determine a cutoff, as discussed below.

Cutoff Determination

As previously mentioned, ATE and NHSTD methods return a list of candidate terms with associated "goodness" scores (calculated in different ways depending on the method). The user must decide where to impose a cutoff. This is done either by fixing the number of candidate terms the method may return (the Top-K method) or by setting a numerical threshold on the "goodness" measure, returning all candidates above it and ignoring the rest.

We took two approaches to determining a cutoff. In the experiments designed to test the HMApp's effectiveness against KeyBERT and TF-IDF, we divided each ranked list into quarters and placed cutoffs after the first, second, and third quarters. This yields three comparison points: terms in the top quarter are considered hateful and the remaining three non-hateful; the top half hateful and the bottom half non-hateful; and the top three quarters hateful and the final quarter non-hateful. Using quarter-based cutoffs rather than numerical thresholds helped us avoid having to identify thresholds of comparable usefulness across the two methods.

In the experiments designed to test the HMApp's expandability, we considered all returned terms hateful and calculated precision on each of the five benchmark datasets. In other words, the only cutoff was the natural one of running out of data, and no terms were classified as non-hateful.

Evaluating the Results at All Possible Cutoffs

As an alternative to selecting a single cutoff for the metrics in Evaluation Metrics, the user can observe the method's performance at every possible cutoff. We consider two such approaches: ROC analysis and precision-recall curves.

ROC Analysis. ROC analysis plots the True Positive (TPR) and False Positive Rates (FPR) across all threshold settings, where TPR is identical to recall as defined above:

$$\text{TPR} = \frac{\text{TP}}{\text{TP}+\text{FN}} = \text{Recall} \qquad \text{FPR} = \frac{\text{FP}}{\text{TN}+\text{FP}}$$

This can be done simultaneously for several methods on the same dataset, yielding multiple curves. Curves whose points sit closest to the northwest corner are better, because the corner corresponds to the optimal result: a TPR of 1 (every positive instance is correctly classified) and an FPR of 0 (no negative instance is misclassified as positive). Consequently, a curve located above another represents a more effective method.

The associated summary statistic is Area Under the Curve (AUC), which, as the name suggests, calculates the geometrical area under the ROC curve. If the curves plotted on the ROC graph do not cross, the AUC is more indicative of performance than when they do.

Precision-Recall Curves. Similar to ROC curves, precision-recall (PR) curves plot precision against recall across all threshold settings, again yielding one curve per method. Here, curves closest to the northeast corner are better, because that corner corresponds to the optimal result: both precision and recall equal 1, meaning every term classified as positive is truly positive, and no positive term is missed. As with ROC analysis, a curve located above another represents a more effective method.

Experimental Design

We ran two series of experiments: effectiveness and expandability.

Effectiveness Experiments

The purpose of the effectiveness experiments is to assess the effectiveness of the HMApp—that is, to show that highly successful tools from adjacent research areas underperform on the newly defined NHSTD task. To this end, we conducted the following experiments on the antisemitism dataset.

The construction of this dataset is described in "Benchmark Data Set Construction." For antisemitism specifically, 10 seed words were used to retrieve posts from 4chan, yielding 10 separate datasets, one per seed word. We ran the HMApp, KeyBERT, and TF-IDF on each dataset, stored the output terms and their goodness scores in descending order, and retained the top 15 from each list.

The top 15 terms output by each system were then labeled using both the hard and soft labeling strategies discussed in "Benchmark Data Set Construction," though only the soft scores are considered here. The 10 lists were merged into a single ranked list per system, yielding three lists of 150 terms: one for the HMApp, one for KeyBERT, and one for TF-IDF.

From these lists, we generated ROC and PR curves and computed precision, recall, and F1 at the three cutoff values mentioned in "Cutoff Determination." These results are shown and discussed in "ROC Analysis and PR Curve Results." We additionally prompted two LLMs to see whether they could match the HMApp's performance. The results are touched upon briefly in "ROC Analysis and PR Curve Results" and reported in full in the Appendix.

Expandability Experiments

For the expandability experiments, all five benchmark datasets were considered, and both hard and soft results are reported. Each dataset was constructed in the same fashion as the antisemitism dataset, using 10 seed words and retaining the top 15 terms per list. Having confirmed the HMApp's effectiveness over KeyBERT and TF-IDF, we ran these experiments only on the HMApp. All results in this section come from 5 lists of 150 terms.

The results fall into two categories: quantitative and qualitative. For the quantitative results, precision is calculated on the full set of 150 terms (all treated as positive classifications), yielding both a hard score and a soft score. For the qualitative results, we examined terms that the HMApp flagged as hateful but that neither web sources nor AI searches confirmed.

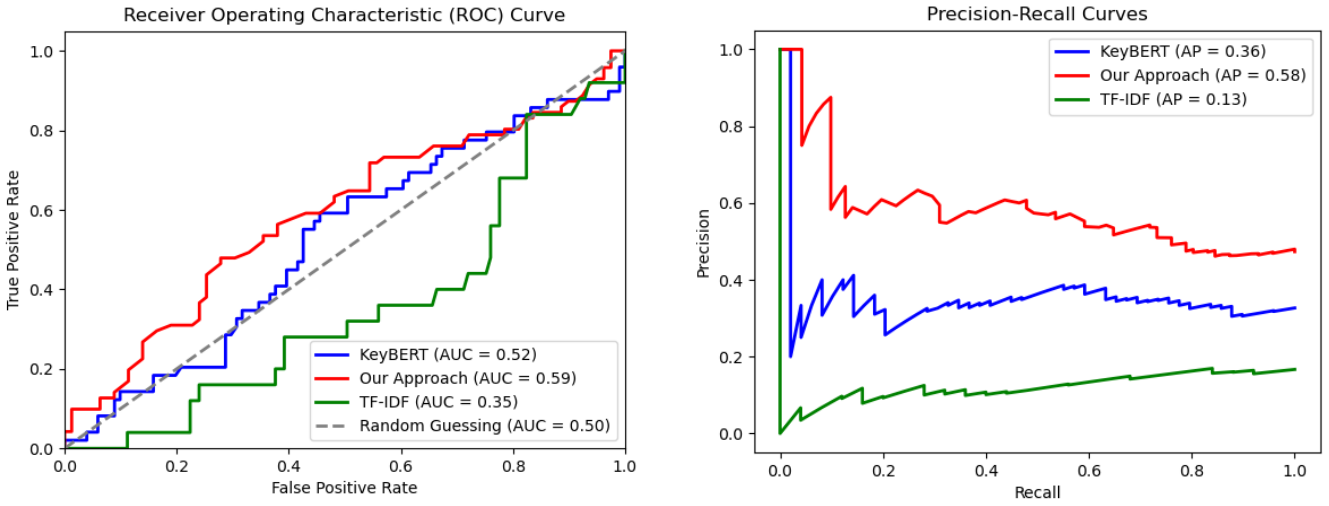


Figure 5. The results of the ROC analysis and PR curve analysis.

Results and Discussion

We now present the results of our experiments, beginning with the effectiveness experiments and concluding with the expandability ones.

Effectiveness Results

ROC Analysis and PR Curve Results

The ROC analysis and the precision-recall curve analysis were performed on the sorted list of terms produced by TF-IDF, KeyBERT, and the HMApp discussed in Effectiveness Experiments, yielding the graphs in Figure 5.

In both graphs, TF-IDF is shown in green, KeyBERT in blue, and the HMApp in red. The ROC graph also includes the "random guessing" line, corresponding to flipping a coin to decide whether or not a term is antisemitic. TF-IDF's performance is clearly the weakest of the three, falling below this line. This supports the hypothesis that a keyword search is not sufficient for this problem, and that semantics needs to be involved, somehow. Indeed, both KeyBERT and the HMApp outperform random guessing on the NHSTD task. Furthermore, the HMApp outperforms KeyBERT, which is unsurprising given that KeyBERT was not designed for the specific NHSTD task while the HMApp was. This demonstrates the need for a new kind of method, as existing tools were not built to search for the novel hateful terminology that hate speech analysts require. In addition to the curves, the AUC values are displayed: 0.35 for TF-IDF, 0.52 for KeyBERT, and 0.59 for HMApp. While the HMApp is again the stronger of the three, both KeyBERT and HMApp are only marginally better than random guessing, suggesting that there is room for improvement.

The PR curve analysis tells a similar story: the red HMApp curve dominates the blue KeyBERT curve, which itself dominates the green TF-IDF curve at every recall point. HMApp's curve is also closer to expectations—precision decreases as recall increases—though the decrease is very slight. This indicates that the precision-recall tradeoff is not a major factor in this problem: recall can be allowed to grow to capture as much terminology as possible without sacrificing too much in precision. That said, HMApp's precision remains quite low (between 0.5 and 0.6), again warranting improvement. The KeyBERT curve is a little strange: between recall values of 0.2 and 0.6, precision increases rather than decreases, suggesting that KeyBERT's retrieval strategy does not follow a clear logic and may not be appropriate for this task. Finally, the TF-IDF curve is subpar and also shows an uncharacteristic simultaneous increase in both precision and recall.

While the two graphs show that the HMApp is superior to KeyBERT and TF-IDF on this task, they also suggest room for improvement. One such improvement could be to replace TF-IDF with KeyBERT, or combine the two, in the search-for-candidate-terms phase of the HMApp. This suggestion stems from the observation that, although KeyBERT's numerical results are on the

weak side, some of the terms it retrieved correctly differed from those found by HMApp, such as "nazi hated freemason."

ROC analysis and PR curves are useful for comparing the overall performance of different methods, but by testing every possible cutoff rather than setting one, they are not a practical approach in deployment. The next section considers experiments where the threshold was fixed at three specific locations.

Precision, Recall, and F-Measure Results

As discussed before, the measure of goodness used by the HMApp is the cosine similarity between the candidate terms and anchor terms, while KeyBERT's is the cosine similarity between the candidate terms and the complete post. TF-IDF, in contrast, ranks terms by their importance to their post relative to all posts in the collection. Since there is no universally optimal threshold for cosine similarity–based tasks or for rankings of the type produced by TF-IDF, we need to experiment with multiple thresholds to identify false negatives. To do so, we divided each sorted list into quarters as discussed in "Evaluation Metrics" and "Cutoff Determination." This allowed us to compute precision, recall, and F1 at each cutoff. The results for TF-IDF, KeyBERT, and the HMApp are shown in Tables 1, 2, and 3.

As expected, since these values correspond to points on the PR curve, the HMApp dominates KeyBERT in both precision and recall at every cutoff, although recall is similar at the second and third cutoffs. Unsurprisingly, TF-IDF's values fall well below those of KeyBERT and HMApp at every cutoff. The F1 values suggest that the last cutoff achieves the best balance between precision and recall. In the next section, we leverage the earlier finding that precision changes little as recall grows, and let recall run all the way to 1.0. The soft score results for antisemitism in Table 4 below confirm this: precision holds steady at 0.5.

Expandability Results

We tested the HMApp's expandability on four additional types of hatred beyond antisemitism. As per "Expandability Experiments," we focused on the precision of all 150 terms retrieved by the app, considering both the hard and soft scores. The results are reported in Table 4 and discussed below.

Table 1. TF-IDF Evaluation for Overall Terms at Different Equal Bins.

Cutoff (top percentile)	Threshold	Precision	Recall	F-Measure
25%	0.36	0.11	0.16	0.13
50%	0.26	0.11	0.32	0.16
75%	0.23	0.13	0.56	0.21

Table 2. KeyBERT Evaluation for Overall Terms at Different Equal Bins.

Cutoff (top percentile)	Threshold	Precision	Recall	F-Measure
25%	0.75	0.27	0.20	0.23
50%	0.69	0.38	0.57	0.46
75%	0.65	0.34	0.78	0.47

Table 3. HMApp Evaluation for Overall Terms at Different Equal Bins.

Cutoff (top percentile)	Threshold	Precision	Recall	F-Measure
25%	0.76	0.59	0.31	0.41
50%	0.72	0.55	0.59	0.57
75%	0.69	0.50	0.79	0.61

Quantitative Results. Table 4 presents the averaged hard and soft precision scores for the five hatred categories. The strongest performance was observed in the anti-LGBTQ+ and anti-Black categories, with hard scores of 0.36 and 0.44, respectively, and high soft scores in both. This suggests that established sources contain a substantial number of well-documented slurs related to these forms of hatred and that the app successfully identifies a considerable portion of them. The notably high soft scores further indicate that the app is effective at retrieving hateful terms not documented in established lexicons.

By contrast, the antisemitism category showed a low hard score of 0.16 but a moderate soft score of 0.50. This pattern suggests that antisemitic expressions are not well documented in formal hate speech lexicons, yet the app still captures a substantial number of antisemitic terms circulating on 4chan. The discrepancy likely reflects the rapid evolution of coded antisemitic language and indirect references, which have yet to be incorporated into established lexicons that tend to be slow to update.

Islamophobia and anti-Asian hatred had two of the lowest hard scores, at 0.16 and 0.11, respectively. Their soft scores were somewhat higher, particularly for anti-Asian hatred, which reached 0.49. This suggests that the language used in these domains is more recent, more context-dependent, or less thoroughly documented in hate speech repositories. Online discourse targeting these groups may also vary significantly across regions and communities, making such terms harder to capture using authoritative sources alone. This underscores the app's usefulness in identifying hateful language that falls outside established lexicons, a pattern that holds across all five categories, where soft scores consistently exceed hard scores and confirm the app's primary value lies in surfacing emergent terminology.[7]

Table 4. The Averaged Hard and Soft Scores for the Five Hatred Categories.

Hatred type	Hard score	Soft score
Antisemitism	0.16	0.50
Anti-Black	0.44	0.63
Islamophobia	0.16	0.25
Anti-Asian	0.11	0.49
Anti-LGBTQ+	0.36	0.63

Qualitative Testing Findings. A qualitative review was also conducted to identify hate-associated terms that did not appear in either established lexicons or web or AI searches. It relied instead on our domain expertise in hate speech. While many of the terms surfaced were borderline depending on context, three stood out as worthy of further investigation: "rootless" in the anti-Black category, and "takfir" and "mushaf" in the Islamophobia category.

The following explains why these terms were retained, though our analysis requires further validation. Although the "rootless" slur usually falls in the antisemitism category, such as in the expression "Rootless Cosmopolitan," it can also be used to slander the Black American community. Particularly, there can be accusations of "Cultural Rootlessness" against that community due to its lost ties to Africa after the slave trade. While "takfir" is not typically used by non-Muslims to slander Muslims, it can be used slanderously within the Muslim community. For example, extremist groups invoke it against any Muslims they deem insufficiently devout (Badar et al., 2017; Baele, 2019; Siegel, 2015). Extremist groups may also distinguish between "pure Muslims" and those they brand as non-believers through takfir. As for "mushaf," though the term itself is not derogatory—it refers to written copies of the Quran—its desecration by burning, tearing, or mishandling it in any other way is considered highly offensive. A post may be considered offensive if the term "mushaf" appears in this context.

Although these terms were not captured by the first two methodologies, they can appear in contexts that carry hateful or derogatory meaning. This step ensures that the evaluation is not limited to what official sources or search retrieval can provide and again underscores the app's value when combined with domain expertise and contextual knowledge.

Limitations

We now present some of the limitations of our research.

Benchmark Data. The only need we had for benchmark data was testing, given the unsupervised nature of the HMApp. Nonetheless, testing is a vital part of the research process, and we needed to gather large and reliable datasets to do it well. Because the problem

7 Note that in the case of Islamophobia, the soft score remains the lowest of all categories of hatred. This may be because the anchor set is very small, containing only 6 terms compared with 16 to 24 in all other categories. Cross-category comparisons should, thus, take this difference into consideration.

we considered is quite subtle, using a Mechanical Turk process was not an option. Instead, we relied on internal resources. As a result, and due to both time and personnel limitations, our benchmark datasets are small—150 labeled terms per dataset. They are reliable in that we closely followed the procedure laid out in "Materials and Methods, Part III: Evaluation," but we had only two coders, whose work did not systematically overlap (though they often discussed areas of uncertainty).

Anchor Terms. Choosing appropriate anchor terms is essential to the success of the HMApp, since they determine the focus of its search. We have not experimented with different subsets of anchor terms or studied the effect of set size.

Design and Parameter Settings. Although we experimented with a number of different settings as discussed in "Materials and Methods, Part II: Methodology," research in this area continues to evolve, and it would be useful to experiment with newer models.

Length of the output. The greatest issue we encountered in practical application is the number of potential terms returned by the HMApp. Given the importance of recall in this work, simply limiting the number of returned terms is not sufficient. We experimented with prompting an LLM about our resulting list, and initial results are encouraging.

Although this issue could be construed as a false-positive problem, it differs from how false positives present in an HSD system. HSD systems issue decisions. The HMApp, instead, returns ranked lists that analysts must sift through. Furthermore, the terms we return are not directly associated with specific posts or people. While the terms can, once selected, be used to find specific posts and the users behind them, this requires an extra step that we expect will be taken only after the analyst has confirmed the term's validity. This, we believe, adds a degree of safety that automated HSD systems do not possess and is sufficient to protect communities and individuals from undue accusations.

Potential mismatch during use. In its current form, the app expects users to enter seed terms related to the kind of hatred they opted for, without the system engaging in any kind of validation. Yet a mismatch will yield poor results because the seed words determine the subset of posts used to search for new terminology. If non-judiciously chosen, they will lead to limited output.

Lack of explanations. The HMApp currently outputs a list of terms and a reliability score, but does not explain how or in what context each term is hateful.

Foreign Languages Tests. The HMApp has been used informally on data outside 4chan with interesting results. It could also be applied to any language, but we have not yet experimented with this. Such experiments would require NLP preprocessing tools adapted to the target language and a list of anchor terms representing the target hatred in that language.

Experiments in practical settings. We have begun experimenting with the HMApp in practical settings. Although our partners are enthusiastic and the early results are encouraging, these experiments have so far been limited in scope.

No Social Study Analysis. While this research contributes a new tool and advances the computer science side of the problem, it does not use this tool to conduct social science analysis. We have, however, begun working with a foundation and hope to produce such an analysis based on our tool's usage in the near future.

Dangers associated with our tool. We acknowledge that our tool can be misused. While we intend it for well-meaning users seeking to use it as part of a process of mitigating hatred on social media, it could be used in the exact opposite way: a malicious actor could identify which terms are already detected by AI tools and invent new ones to evade them. We aim to mitigate this problem by distributing the tool sparingly and in a controlled manner, though we recognize that this may fail and could cause added harm.

Conclusion

Summary

This study presents HMApp, an innovative app that can be used as part of the hate speech mitigation process. Its purpose is to identify novel expressions of hatred early, in the hope of stymying its spread—and the spread of hatred along with them. Although related to both HSD and ATE tools, the HMApp addresses a problem that neither fully solves, though we recognize that certain ATE tools come close.

Our contributions are twofold. In addition to proposing a new NLP task, we propose a powerful framework and versatile search criterion able to focus an otherwise unsupervised search process that addresses that task. The framework extends easily to different kinds of hatred and, with appropriate pre-processing, could be applied to other languages as well, demonstrating the power of lightly guided unsupervised search.

To assess HMApp's effectiveness, we conduct a thorough evaluation, including comparisons against one of the foremost ATE tools available, a tool that uses underlying technologies similar to the HMApp's, and another that relies on an older, purely statistical strategy. Our experiments show that the HMApp addresses our particular problem more effectively than either ATE tool.

An important aspect of the HMApp is that it is built on the divide-and-conquer principle. Unlike many HSD tools that attempt to bundle all forms of hatred into a single system, we strongly believe that each type's unique expressions warrant separate treatment. Nonetheless, we wanted a tool capable of addressing all forms of hatred, and the HMApp's modularity makes this possible. Its architecture allows it to transition seamlessly from one type of hatred to another. Indeed, through its use of "anchor terms," a small set of known terms representative of the particular kind of hatred we want to target, the HMApp can easily be fitted to the speech in question.

We demonstrate this expandability by experimenting on five types of hatred: antisemitism, anti-Black hatred, Islamophobia, anti-Asian hatred, and anti-LGBTQ+ hatred. By comparing the app's ability to retrieve terms already listed in lexicons with its ability to identify hateful terminology beyond them, we show that the HMApp can catch novel expressions of hatred. It performs very well on anti-Black and anti-LGBTQ+ hatred, moderately well on anti-Asian and antisemitic hatred, but more poorly on Islamophobia.

Future Work

We are pursuing several avenues for future research: understanding the observed variation in performance, improving the app's current functionality, adding new features, and conducting user studies with researchers and practitioners working in hate speech monitoring and intervention.

To account for variation in performance, we propose studying the nature of hatred expressed against different communities. In particular, we will explore the degree and type of covertness used in hateful posts targeting each community and analyze which anchor terms best focus the HMApp's search in each case. We acknowledge that anchor terms will not always suffice, as the expression of hatred can go beyond terminology and the concepts a term and its context entail. As in Mendelsohn et al. (2023) and Su et al. (2026), we may need to uncover more complex and subtle relationships among concepts to understand the hatred expressed in certain posts.

To improve current functionality, we plan to experiment with more sophisticated candidate keyword extraction methods and embedding models beyond BERT, as well as test different subsets of anchor terms to improve both precision and recall. An additional line of improvement involves streaming data from social media platforms beyond 4chan. Because many social media platforms have either closed or monetized their APIs, it can be difficult to obtain real-time data at scale. However, we are currently exploring alternative data collection methods to address this limitation.

For new functionality, we plan to expand the app to cover additional types of hatred. Future additions could include misogyny, hatred against the Hispanic and Latino community, Native Americans, and other forms of targeted hatred. We will also turn our attention to hatred in foreign languages.

We also plan to incorporate an internal classification system to ensure that posts under consideration are genuinely hateful and represent the specific type of hatred being examined. This is particularly important as we expand to platforms beyond 4chan's /pol/ board, where many posts do not convey hatred at all. Distinguishing between hateful and non-hateful content, and the ethical issues this entails, will then become essential.

Another planned addition is the explainability unit. Surfacing coded or emergent terminology is valuable, but understanding why these terms were flagged would be more so. Such explainability would also serve as a self-validation mechanism, improving the tool's overall performance.

Lastly, user studies, which we are exploring in parallel, will bring the app one step closer to deployment. We are currently testing the app's usefulness and usability with a cohort of students and working closely with one foundation's social media hate speech discovery team. The next phase will bring in more researchers and practitioners who specialize in hate speech. Our hope is that this app will ultimately support those working to combat hatred and discrimination on social media.

Funding Disclosure
The app, as described in the paper, was supported in part by an American University Signature Research Initiative grant.

Conflicts of Interest
None to declare.

Relevant Institutional or Advisory Roles
None to declare.

AI/LLM Disclosure
We used Claude Opus 4.6 and 4.7 for grammar, style editing, and rewording select sentences for clarity and flow. Grammarly, an AI-powered writing assistant, was also used to identify typos and grammatical errors.

Data and Code Availability Statement
The social media posts used in this paper's analysis are available upon request. Due to the sensitive nature of the data, which contains hateful content directed at protected groups, we use a request approach to ensure they come from researchers, practitioners, policymakers, and other parties with a legitimate interest in studying or mitigating online hate speech.

The link to our GitHub repository will be: https://github.com/patrickywu/EmergentHateSpeech-HMApp. At the moment, there is no content in the repository. We will vet the code and push it to the repository soon.

The app itself is currently accessible upon request. We are actively developing a plan for broader release, but as discussed in Limitations, the tool carries a risk of misuse. For example, a malicious actor could use it to identify terms currently detected by AI moderation systems and craft new ones to evade them. Thus, we are proceeding cautiously and limiting access while we work out a distribution mechanism that balances scholarly and practitioner access against the risk of misuse.

References
Almuqati, M. T., Sidi, F., Mohd Rum, S. N., Zolkepli, M., & Ishak, I. (2024). Challenges in supervised and unsupervised learning: A comprehensive overview. International Journal on Advanced Science, Engineering & Information Technology, 14(4), 1449–1455. https://doi.org/10.18517/ijaseit.14.4.20191

Anti-Defamation League. (2024). Online hate and harassment: The American experience 2024 [Report]. ADL Center for Technology & Society. https://www.adl.org/sites/default/files/documents/2024-06/online-hate-and-harassment-the-american-experience-v2024.pdf

Arcila Calderón, C., Sánchez Holgado, P., Gómez, J., Barbosa, M., Qi, H., Matilla, A., Amado, P., Guzmán, A., López-Matías, D., & Fernández-Villazala, T. (2024). From online hate speech to offline hate crime: The role of inflammatory language in forecasting violence against migrant and LGBT communities. Humanities and Social Sciences Communications, 11(1), 1369. https://doi.org/10.1057/s41599-024-03899-1

Assimakopoulos, S., & Serafis, D. (2025). Why soft hate speech matters: Argumentativity and the dispersion of hatred towards minorities. Critical Discourse Studies, 23(3), 347–366. https://doi.org/10.1080/17405904.2024.2440869

Badar, M., Nagata, M., & Tueni, T. (2017). The radical application of the Islamist concept of takfir. Arab Law Quarterly, 31(2), 134–162. https://doi.org/10.1163/15730255-31020044

Baele, S. J. (2019). Conspiratorial narratives in violent political actors' language. Journal of Language and Social Psychology, 38(5–6), 706–734. https://doi.org/10.1177/0261927X19868494

Becker, M. J., & Troschke, H. (2023). Decoding implicit hate speech: The example of antisemitism. In C. Strippel, S. Paasch-Colberg, M. Emmer, & J. Trebbe (Eds.), Challenges and perspectives of hate speech research (Vol. 12, pp. 335–352). https://doi.org/10.48541/dcr.v12.20

Bellinger, C., Sharma, S., Zaïane, O. R., & Japkowicz, N. (2017). Sampling a longer life: Binary versus one-class classification revisited. First International Workshop on Learning with Imbalanced Domains: Theory and Applications, 64–78.

Bennani-Smires, K., Musat, C., Hossmann, A., Baeriswyl, M., & Jaggi, M. (2018). Simple unsupervised keyphrase extraction using sentence embeddings. Proceedings of the 22nd Conference on Computational Natural Language Learning, 221–229.

Bermudez-Villalva, A., Mehrnezhad, M., & Toreini, E. (2025). Measuring online hate on 4chan using pre-trained deep learning models. IEEE Transactions on Technology and Society, 6(2), 200–209. https://doi.org/10.1109/TTS.2025.3549931

Binder, J. F., & Kenyon, J. (2022). Terrorism and the internet: How dangerous is online radicalization? Frontiers in Psychology, 13:997390 https://doi.org/10.3389/fpsyg.2022.997390

Blackington, C., & Cayton, F. (2025). To dog-whistle or to bark? Elite communication strategies when invoking conspiracy theories. Government and Opposition, 60(2), 382–403. https://doi.org/10.1017/gov.2024.21

Caselli, T., Basile, V., Mitrović, J., & Granitzer, M. (2021). HateBERT: Retraining BERT for abusive language detection in English. In A. Mostafazadeh Davani, D. Kiela, M. Lambert, B. Vidgen, V. Prabhakaran, & Z. Waseem (Eds.), Proceedings of the 5th workshop on online abuse and harms (WOAH 2021) (pp. 17–25). Association for Computational Linguistics. https://doi.org/10.18653/v1/2021.woah-1.3

Chaudhary, H. P., & Asarsa, J. (2025). The influence of social media and digital communication on language use and evolution. International Journal of Scientific Research in Humanities and Social Sciences, 2(2), 1–4. https://doi.org/10.32628/IJSRHSS25221

Choudhary, D. D., Reddy, M. S., & Kumar, M. A. (2026). Hate speech sentiment detection with topic modeling-based feature engineering. In C. Modi, G. K. Verma, V. Thenkanidiyoor, & L. Brankovic (Eds.), Machine learning, image processing, network security and data sciences: 6th International Conference, MIND 2024, Goa, India, December 20–21, 2024, revised selected papers, Part I (pp. 305–316). Springer.

Das, S., Dutta, A., Roy, K., Mondal, A., & Mukhopadhyay, A. (2024). A survey on automatic online hate speech detection in low-resource languages. arXiv. https://arxiv.org/abs/2411.19017

Davidson, T., Warmsley, D., Macy, M., & Weber, I. (2017). Automated hate speech detection and the problem of offensive language. Proceedings of the International AAAI Conference on Web and Social Media, 11(1), 512–515. https://doi.org/10.1609/icwsm.v11i1.14955

Delgado, R., & Stefancic, J. (1995). Ten arguments against hate-speech regulation: How valid? Northern Kentucky Law Review, 23(3), 475. https://scholarship.law.ua.edu/fac_articles/564

Di Marco, N., Loru, E., Bonetti, A., Serra, A. O. G., Cinelli, M., & Quattrociocchi, W. (2024). The evolution of language in social media comments. arXiv. https://arxiv.org/abs/2406.11450

Dubois, E., & Reepschlager, A. (2024). How harassment and hate speech policies have changed over time: Comparing Facebook, Twitter and Reddit (2005–2020). Policy & Internet, 16(3), 523–542. https://doi.org/10.1002/poi3.387

European Commission against Racism and Intolerance. (2015). ECRI general policy recommendation no. 15 on combating hate speech. Council of Europe.

Fasching, N., & Lelkes, Y. (2025). Model-dependent moderation: Inconsistencies in hate speech detection across LLM-based systems. In W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), Findings of the association for computational linguistics: ACL 2025 (pp. 22271–22285). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.findings-acl.1144

Fortuna, P., & Nunes, S. (2018). A survey on automatic detection of hate speech in text. ACM Computing Surveys (CSUR), 51(4), 1–30.

Gandhi, A., Ahir, P., Adhvaryu, K., Shah, P., Lohiya, R., Cambria, E., Poria, S., & Hussain, A. (2024). Hate speech detection: A comprehensive review of recent works. Expert Systems, 41(8), e13562.

Garcia, C. M., Abilio, R., Koerich, A. L., de Souza Britto, A., Jr., & Barddal, J. P. (2025). Concept drift adaptation in text stream mining settings: A systematic review. ACM Transactions on Intelligent Systems and Technology, 16(2). https://doi.org/10.1145/3704922

GLAAD. (2024). Guide to anti-LGBTQ online hate and disinformation. GLAAD Social Media Safety Program. https://glaad.org/smsi/anti-lgbtq-online-hate-speech-disinformation-guide/

Goel, V., Sahnan, D., Dutta, S., Bandhakavi, A., & Chakraborty, T. (2023). Hatemongers ride on echo chambers to escalate hate speech diffusion. PNAS Nexus, 2(3), pgad041. https://doi.org/10.1093/pnasnexus/pgad041

Grootendorst, M. (2020). KeyBERT: Minimal keyword extraction with BERT. (Version v0.3.0). Zenodo. https://doi.org/10.5281/zenodo.4461265

He, B., Ziems, C., Soni, S., Ramakrishnan, N., Yang, D., & Kumar, S. (2022). Racism is a virus: Anti-Asian hate and counterspeech in social media during the COVID-19 crisis. Proceedings of the 2021 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining, ASONAM '21, 90–94. https://doi.org/10.1145/3487351.3488324

Hine, G., Onaolapo, J., De Cristofaro, E., Kourtellis, N., Leontiadis, I., Samaras, R., Stringhini, G., & Blackburn, J. (2017). Kek, cucks, and god emperor Trump: A measurement study of 4chan's politically incorrect forum and its effects on the web. Proceedings of the International AAAI Conference on Web and Social Media, 11(1), 92–101. https://doi.org/10.1609/icwsm.v11i1.14893

Hoeken, S., Spliethoff, S., Schwandt, S., Zarrieß, S., & Alacam, Ö. (2023). Towards detecting lexical change of hate speech in historical data. In N. Tahmasebi, S. Montariol, H. Dubossarsky, A. Kutuzov, S. Hengchen, D. Alfter, F. Periti, & P. Cassotti (Eds.), Proceedings of the 4th workshop on computational approaches to historical language change (pp. 100–111). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.lchange-1.11

Hoeken, S., Zarrieß, S., & Alacam, Ö. (2024). Hateful word in context classification. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), Proceedings of the 2024 conference on empirical methods in natural language processing (pp. 172–186). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-main.10

Horta Ribeiro, M., Cheng, J., & West, R. (2023). Automated content moderation increases adherence to community guidelines. Proceedings of the ACM Web Conference 2023, WWW '23, 2666–2676. https://doi.org/10.1145/3543507.3583275

Hswen, Y., Xu, X., Hing, A., Hawkins, J. B., Brownstein, J. S., & Gee, G. C. (2021). Association of "#COVID19" versus "#chinesevirus" with anti-Asian sentiments on Twitter: March 9–23, 2020. American Journal of Public Health, 111(5), 956–964. https://doi.org/10.2105/AJPH.2021.306154

Jhaver, S., Birman, I., Gilbert, E., & Bruckman, A. (2019). Human-machine collaboration for content regulation: The case of Reddit automoderator. ACM Transactions on Computer-Human Interaction, 26(5), 1–35. https://doi.org/10.1145/3338243

Johnson, D. G., Mattan, B. D., Flores, N., Lauharatanahirun, N., & Falk, E. B. (2022). Social-cognitive and affective antecedents of code switching and the consequences of linguistic racism for Black people and people of color. Affective Science, 3(1), 5–13. https://doi.org/10.1007/s42761-021-00072-8

Kageura, K., & Umino, B. (1996). Methods of automatic term recognition: A review. Terminology, 3(2), 259–289.

Kalfas, N. K. (2025). When words become weapons: How hate speech threatens democracy. International Institute for Democracy and Electoral Assistance (International IDEA). https://www.idea.int/news/when-words-become-weapons-how-hate-speech-threatens-democracy

Kikkisetti, D., Mustafa, R., Melillo, W., Corizzo, R., Boukouvalas, Z., Gill, J., & Japkowicz, N. (2024). Coded term discovery for online hate speech detection. 2024 IEEE 11th International Conference on Data Science and Advanced Analytics (DSAA), 1-10. https://doi.org/10.1109/DSAA61799.2024.10722816

King, G., Lam, P., & Roberts, M. E. (2017). Computer-assisted keyword and document set discovery from unstructured text. American Journal of Political Science, 61(4), 971–988. https://doi.org/10.1111/ajps.12291

Kiritchenko, S., Nejadgholi, I., & Fraser, K. C. (2021). Confronting abusive language online: A survey from the ethical and human rights perspective. Journal of Artificial Intelligence Research, 71, 431–478.

Kolla, M., Salunkhe, S., Chandrasekharan, E., & Saha, K. (2024). LLM-mod: Can large language models assist content moderation? Extended Abstracts of the CHI Conference on Human Factors in Computing Systems, CHI EA '24. https://doi.org/10.1145/3613905.3650828

Kumar, S., Nagar, A., Kumar, A., & Singh, A. (2022). Hate speech detection: A survey. 2022 4th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N), 171–176.

Laub, Z. (2019). Hate speech on social media: Global comparisons. Council on Foreign Relations Backgrounder. https://www.cfr.org/backgrounders/hate-speech-social-media-global-comparisons

Lillian, D. L. (2007). A thorn by any other name: Sexist discourse as hate speech. Discourse & Society, 18(6), 719–740. https://doi.org/10.1177/0957926507082193

Lupu, Y., Sear, R., Velásquez, N., Leahy, R., Restrepo, N. J., Goldberg, B., & Johnson, N. F. (2023). Offline events and online hate. PLOS ONE, 18(1), 1–14. https://doi.org/10.1371/journal.pone.0278511

Madhu, S. (2025). Tracing online hate long-term: Using machine learning to connect Twitter usage per year to its hate speech. Critical Debates in Humanities, Science and Global Justice, 5(2), 148–158. https://criticaldebateshsgj.scholasticahq.com/article/141456-tracing-online-hate-long-term-using-machine-learning-to-connect-twitter-usage-per-year-to-its-hate-speech

Magu, R., & Luo, J. (2018). Determining code words in euphemistic hate speech using word embedding networks. In D. Fišer, R. Huang, V. Prabhakaran, R. Voigt, Z. Waseem, & J. Wernimont (Eds.), Proceedings of the 2nd workshop on abusive language online (ALW2) (pp. 93–100). Association for Computational Linguistics. https://doi.org/10.18653/v1/W18-5112

Malik, B., Iftikhar, F., & Gurmani, T. (2025). Transformative impact of social media platforms on language evolution: Creation and adoption of emerging lexicon. Journal of Arts and Linguistics Studies, 3, 243–267. https://doi.org/10.71281/jals.v3i1.218

Melillo, W., Emami, J., Guarinos, S., Kikkisetti, D., Klein, M., Liubovich, L., Mustafa, R. U., & Japkowicz, N. (2024). Seeking optimal human/machine collaborative practice in antisemitic terminology detection. 2024 IEEE Digital Platforms and Societal Harms (DPSH), 1–9. https://doi.org/10.1109/DPSH60098.2024.10775254

Mendelsohn, J., Le Bras, R., Choi, Y., & Sap, M. (2023). From dogwhistles to bullhorns: Unveiling coded rhetoric with language models. In A. Rogers, J. Boyd-Graber, & N. Okazaki (Eds.), Proceedings of the 61st annual meeting of the association for computational

linguistics (volume 1: Long papers) (pp. 15162–15180). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.acl-long.845

Meta. (2022). Facebook community standards: Hate speech. https://transparency.meta.com/policies/community-standards/hate-speech/.

Mustafa, R. U., Dupart, R., Smith, G., Ashraf, N., & Japkowicz, N. (2025). Analyzing Islamophobic discourse using semi-coded terms and LLMs. arXiv. https://arxiv.org/abs/2503.18273

Nadim, M., Akopian, D., & Matamoros, A. (2023). A comparative assessment of unsupervised keyword extraction tools. IEEE Access, 11, 144778–144798. https://doi.org/10.1109/ACCESS.2023.3344032

Nielsen, L. B. (2002). Subtle, pervasive, harmful: Racist and sexist remarks in public as hate speech. Journal of Social Issues, 58(2), 265–280. https://doi.org/10.1111/1540-4560.00260

Ocampo, N. B., Sviridova, E., Cabrio, E., & Villata, S. (2023). An in-depth analysis of implicit and subtle hate speech messages. In A. Vlachos & I. Augenstein (Eds.), Proceedings of the 17th conference of the European chapter of the association for computational linguistics (pp. 1997–2013). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.eacl-main.147

Olteanu, A., Castillo, C., Boy, J., & Varshney, K. (2018). The effect of extremist violence on hateful speech online. Proceedings of the International AAAI Conference on Web and Social Media, 12(1). https://doi.org/10.1609/icwsm.v12i1.15040

Parker, S., & Ruths, D. (2023). Is hate speech detection the solution the world wants? Proceedings of the National Academy of Sciences, 120(10), e2209384120. https://doi.org/10.1073/pnas.2209384120

Paz, M. A., Montero-Díaz, J., & Moreno-Delgado, A. (2020). Hate speech: A systematized review. Sage Open, 10(4), 2158244020973022. https://doi.org/10.1177/2158244020973022

Polignano, M., Colavito, G., Musto, C., de Demmis, M., & Semeraro, G. (2022). Lexicon enriched hybrid hate speech detection with human-centered explanations. Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization, UMAP '22 Adjunct, 184–191. https://doi.org/10.1145/3511047.3537688

Rieger, D., Kümpel, A. S., Wich, M., Kiening, T., & Groh, G. (2021). Assessing the extent and types of hate speech in fringe communities: A case study of alt-right communities on 8chan, 4chan, and Reddit. Social Media + Society, 7(4), 20563051211052906. https://doi.org/10.1177/20563051211052906

Rosen, Z. P., & Walther, J. B. (2025). Social processes in the intensification of online hate: The effects of verbal replies to anti-Muslim and anti-Jewish posts following 7 October 2023. Social Media + Society, 11(4), 20563051251383635. https://doi.org/10.1177/20563051251383635

Saha, K., Chandrasekharan, E., & De Choudhury, M. (2019). Prevalence and psychological effects of hateful speech in online college communities. Proceedings of the 10th ACM Conference on Web Science, WebSci '19, 255–264. https://doi.org/10.1145/3292522.3326032

Šajatović, A., Buljan, M., Šnajder, J., & Bašić, B. D. (2019). Evaluating automatic term extraction methods on individual documents. Proceedings of the Joint Workshop on Multiword Expressions and WordNet (MWE-WN 2019), 149–154.

Salton, G., & McGill, M. (1986). Introduction to Modern Information Retrieval. McGraw-Hill.

Schmidt, A., & Wiegand, M. (2017). A survey on hate speech detection using natural language processing. Proceedings of the Fifth International Workshop on Natural Language Processing for Social Media, 1–10.

Sharma, P., & Li, Y. (2019). Self-supervised contextual keyword and keyphrase retrieval with self-labelling. Preprints. https://doi.org/10.20944/preprints201908.0073.v1

Shmulewitz, D., Levitin, M. D., Skvirsky, V., Vider, M., Lev-Ran, S., & Mikulincer, M. (2025). Exposure to online hate speech is positively associated with

post-traumatic stress disorder symptom severity. Scientific Reports, 15, 29869. https://doi.org/10.1038/s41598-025-16168-1

Siegel, A. (2015). Sectarian Twitter wars: Sunni-Shia conflict and cooperation in the digital age [Carnegie Paper]. Carnegie Endowment for International Peace. https://carnegieendowment.org/research/2015/12/sectarian-twitter-wars-sunni-shia-conflict-and-cooperation-in-the-digital-age

Su, X., Inkpen, D., & Japkowicz, N. (2026). SoftHate-Bench: Evaluating moderation models against reasoning-driven, policy-compliant hostility. Proceedings of the ACM Web Conference 2026, 4862–4873.

Subramanian, M., Sathiskumar, V. E., Deepalakshmi, G., Cho, J., & Manikandan, G. (2023). A survey on hate speech detection and sentiment analysis using machine learning and deep learning models. Alexandria Engineering Journal, 80, 110–121.

Susanto, A. K., Amin, F. H., & Sudding, M. F. J. (2025). Social media and the evolution of English: A linguistic study of online discourse and language change. Journal Scientific of Mandalika, 6(8), 3359–3366. https://ojs.cahayamandalika.com/index.php/jomla/article/view/4722

Taylor, J., Peignon, M., & Chen, Y.-S. (2017). Surfacing contextual hate speech words within social media. arXiv. https://arxiv.org/abs/1711.10093

Tillyris, D. (2024). Dog-whistling and democracy. The British Journal of Politics and International Relations, 26(4),
1015–1032. https://doi.org/10.1177/13691481231208147

Tran, H. T. H., Martinc, M., Caporusso, J., Delaunay, J., Doucet, A., & Pollak, S. (2026). Recent advances in automatic term extraction: A comprehensive survey. ACM Computing Surveys, 58(9), Article 226. https://doi.org/10.1145/3787584

Tsesis, A. (2002). Destructive messages: How hate speech paves the way for harmful social movements. NYU Press. https://muse.jhu.edu/pub/193/monograph/book/
10922

United Nations Office on Genocide Prevention and the Responsibility to Protect. (2023). Countering and addressing online hate speech: A guide for policy makers and practitioners (p. 20). United Nations. https://digitallibrary.un.org/record/4064117

Vargas, F., Carvalho, I., Pardo, T. A., & Benevenuto, F. (2025). Context-aware and expert data resources for Brazilian Portuguese hate speech detection. Natural Language Processing, 31(2), 435–456.

Vargas, F., Rodrigues de Góes, F., Carvalho, I., Benevenuto, F., & Pardo, T. (2021). Contextual-lexicon approach for abusive language detection. In R. Mitkov & G. Angelova (Eds.), Proceedings of the international conference on recent advances in natural language processing (RANLP 2021) (pp. 1438–1447). INCOMA Ltd. https://aclanthology.org/2021.ranlp-1.161/

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, & R. Garnett (Eds.), Advances in neural information processing systems (Vol. 30). Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf

Walther, J. B. (2022). Social media and online hate. Current Opinion in Psychology, 45, 101298. https://doi.org/10.1016/j.copsyc.2021.12.010

Williams, M. (2019). The connection between online hate speech and real-world hate crime. OUPblog, Oxford University Press. https://blog.oup.com/2019/10/connection-between-online-hate-speech-real-world-hate-crime/

Xu, K., Feng, Y., Li, Q., Dong, Z., & Wei, J. (2025). Survey on terminology extraction from texts. Journal of Big Data, 12(1), 29.

Zampieri, M., Malmasi, S., Nakov, P., Rosenthal, S., Farra, N., & Kumar, R. (2019). Predicting the type and target of offensive posts in social media. Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), 1415–1420.

Zannettou, S., Caulfield, T., Blackburn, J., De Cristofaro, E., Sirivianos, M., Stringhini, G., & Suarez-Tangil, G. (2018). On the origins of memes by means of fringe web communities. Proceedings of the Internet Measurement Conference 2018, IMC '18, 188–202. https://doi.org/10.1145/3278532.3278550

Zannettou, S., Caulfield, T., De Cristofaro, E., Kourtellis, N., Leontiadis, I., Sirivianos, M., Stringhini, G., & Blackburn, J. (2017). The web centipede: Understanding how web communities influence each other through the lens of mainstream and alternative news sources. Proceedings of the 2017 Internet Measurement Conference, IMC '17, 405–417. https://doi.org/10.1145/3131365.3131390

Appendix A

Full List of Anchor Terms for Each Type of Hatred

Reader discretion is advised here for strong and offensive language.

> The list of anchor terms used for anti-Black hatred is: nigger / niger / nig / nigor / nigra / nigar / niggur / nigga / niggah / niggar / nigguh / niggress / nigette / negro / neger, Jiggaboo / jiggerboo / niggerboo / jiggabo / jigarooni / jijjiboo, cotton picker, moon cricket, darky / darkey / darkie, pickaninny, porch monkey, groid, jungle bunny, sambo, Uncle Tom, golliwog, spear chucker, alligator bait / 'gator bait, nig-nog / nog / nignog, coon / cooney, buckwheat, spook, ape, Jim Crow / Crow, spade, and blackie.

> The list of anchor terms used for antisemitism is: kike, blood Libel, the Goyim know, Great Replacement, Holocaust/gas chambers, Holohoax, Jew coup, Jew down, Jewish feature, Jewish figure, Jewish lighting, Jewish power, Kosher tax, laughing Jew, Nazi symbols, protocols of the elders of Zion, QAnon, smirking merchant, Zionism is racism, and Zionist Occupied Government.

> The list of anchor terms used for anti-Asian hatred is the following (again, reader discretion is advised for strong and offensive language): slant eye, Chinaman, yellow Asian, banana, twinkie, ching chong, chink / chinky, chonky, coolie, curry-muncher, zipper head, Pajeet, yellow invader, rice nigger, Mongoloid, and gook.

> The list of anchor terms used for anti-LGBTQ+ hatred is: faggot / fagot / faggotry / faggoty / fag / faggy / fagged, homo, tranny, swishy / swisher, dyke / bull-dyke / bulldagger, twink, chaser, gay lord / gaylord, fairy, daisy / pansy / tulip, cornhole / cornholer, flamer / flaming, alphabet people, fruit / fruity, fag / hag, friend of Dorothy, Nancy boy, lipstick lesbian, batting for the other team, punker, capon, bean queen, curry queen, and Harry Hoofter.

> The list of anchor terms used for Islamophobia is: Muzat, Muzzies, Plslam, Mohammedan, Mudslime, and Turk.

Outputs from Generative LLMs

We experimented with two different AI Systems—Perplexity AI and Google's Gemini—to test their ability to retrieve emerging hateful terminology. Our experiments show that they are not aware of the specific novel, emerging and subtle terms that our app is able to extract. On the other hand, it is interesting to note that Gemini extracts terms that our app may not have access to due to the default source it consults—4chan /pol/: antisemitic terms related to Israel and Zionism. That is because the users of the default source are not that concerned about that aspect of antisemitism, relying more on the traditional kind of antisemitism. If a user were to input a database containing posts related to anti-Israel content and query it with related seed terms, then we believe that the app would retrieve such content and point to more subtle terminology as well.

We used Perplexity on November 14, 2025 with the following prompt to test whether it can retrieve emerging antisemitic terminology: "list emerging antisemitic terminology". The LLM's response was as follows:

Emerging antisemitic terminology often includes both newly coined slurs and coded language, alongside terms that have been revived or recontextualized in recent years to spread hate discreetly on social and digital platforms. Here is a current list of notable examples:

Recent Coded and Direct Terms

- Cabal: Used to refer to an alleged secretive Jewish conspiracy.
- Cosmopolitan Elite: Dog whistle suggesting Jewish dominance in finance or culture.
- Cultural Marxism: Blaming Jews for perceived destructive cultural changes.
- Deicide: Referring to Jews as "killers of Christ," a historic source of hatred.
- The Goyim Know: Used to express concern that non-Jews are learning about supposed Jewish plans.

- Holocough/Holohoax/Holocaust fake: Terms used to deny, mock, or delegitimize the Holocaust.
- Jewish Lobby/Zionist Occupied Government (ZOG): Suggesting undue Jewish influence over political systems.
- Not the Real Jews/Khazar Jew: Denying Jewish identity or legitimacy by claiming Jews are "fake".
- Rothschild, Soros: Used to indicate supposed Jewish control of global events or finances.
- Jew Down/Jewish Capitalist/Jewish Communist: Attributing negative traits or global control to Jewish people.
- IHRA: Sometimes weaponized against Jews by misusing definitions during debates on antisemitism.
- Blood Libel: Revived accusations relating to age-old myths about Jews.

Emerging and Trending Slurs on Social Media

- Gas chamber/gas Jews
- Bankroll/Jewish banks
- Elite/establishment
- Heeb, Joo, Kike (classic slurs revived in coded memes or digital formats)
- Holo (shortened from Holocaust, used in meme formats)
- Jew horns/Jew nose/hook nose (physical stereotype references)
- Khazar/Kazhar Jew
- Jewish money/Jewishness

Notes on Context and Adaptation

Many of these terms can appear in disguised or coded forms, designed to avoid detection or moderation on public platforms. They frequently overlap with anti-Zionist rhetoric, especially when criticism of Israel or its policies crosses into generalizations about Jews as a group. Experts continue to monitor new emerging terminology, especially as hate groups update their language faster than public awareness can follow.

We repeated our experiment on Gemini on the same day, using the same prompt, and obtained the following response from the LLM:

Identifying emerging antisemitic terminology often involves recognizing how classic antisemitic tropes are repackaged into new phrases, metaphors, or "dog whistles," particularly in online spaces.

Here is a list of contemporary and emerging antisemitic terminology, phrases, and concepts, often used to subtly or overtly perpetuate hate:

1. Antisemitic Dog Whistles and Coded Language

These terms often appear innocuous but are understood by antisemitic groups to refer to Jews controlling global affairs or finances.

- (((Echo))): The use of three sets of parentheses around a name, often on social media, to signify or "echo" that the person is Jewish.
- Globalist or Globalists: An updated term often used as a code word for Jews, implying they have no loyalty to their country and instead seek to control the world through international systems (a contemporary version of the "Jewish world conspiracy" trope).
- Cosmopolitan Elite: Similar to "globalist," this phrase is used to accuse Jews of being a disloyal, wealthy, and powerful class that is detrimental to national interests.
- The Goyim Know / The Noticing: A phrase and concept originating online used to signify that "non-Jews" (Goyim) are starting to "notice" the alleged Jewish conspiracy to control various aspects of society (media, government, etc.).
- Holohoax / Holocough: Terms used to deny or trivialize the Holocaust. "Holocough" specifically emerged during the COVID-19 pandemic.

- Anudda Shoah: A phrase designed to mock Jewish people by suggesting they inappropriately or excessively invoke the Holocaust ("another Shoah").

2. Terms Relating to Israel and Zionism (When used in an Antisemitic Context)

While criticism of the State of Israel's policies is not inherently antisemitic, certain phrases and accusations cross the line into antisemitism when they: deny the Jewish right to self-determination, apply a double standard to Israel, or use classic antisemitic imagery/tropes to describe Israel or Israelis.

- "From the River to the Sea": While a political slogan with various interpretations, when used in an antisemitic context, the Anti-Defamation League (ADL) notes it is a charge denying the Jewish right to self-determination and often implies the violent removal of Jews from their ancestral homeland.
- "Zio" / Zionist Occupied Government (ZOG): "Zio" is a derogatory shortening of "Zionist." ZOG is a white supremacist conspiracy theory alleging that Jews secretly control a Western government.
- Settler Colonialist: When this label is used to universally deny the Jewish people's historical connection to the Land of Israel, it functions as an antisemitic trope by portraying Jews as foreign invaders and delegitimizing the existence of the state.
- Deadly Exchange: A campaign and term used to imply that the training of U.S. police forces by Israeli counterparts promotes police brutality, often using classic tropes of Jewish malice and control.
- Watermelon Symbol: Recently, in specific online contexts related to the Israeli-Palestinian conflict, the use of the watermelon emoji has operated as a dog whistle for a broader ideological stance that denies Israel's right to exist.

3. Online Memes and Imagery

Antisemitic tropes are frequently spread through coded imagery and memes online:

- Happy Merchant: A grotesque, widely circulated online caricature of a Jewish man with exaggerated features, depicted as greedy and conniving.
- Pepe the Frog (in certain contexts): While originally a non-political meme, it has been co-opted by white supremacists and antisemites who use variations of it, sometimes alongside the "Happy Merchant," to promote hateful messages.

The terminology of hate is constantly evolving, particularly online, as extremists seek to bypass moderation and spread their messages.

Would you like to search for resources on how to report antisemitic language you encounter online?

Additional Results

Table A1 contains the soft precision, recall, and F1 metrics for all the different hate categories at different thresholds. Table A2 contains the recall, precision, and F1 score at progressively lower thresholds; terms were selected that were just under the given threshold.

Table A1. Soft Scoring Metrics by Hate Category and Threshold.

Hate	Threshold	Soft precision	Soft recall	Soft F_1
Anti-Asian	0.76	0.92	0.47	0.62
Anti-Asian	0.75	0.68	0.70	0.69
Anti-Asian	0.72	0.59	0.85	0.70
Islamophobia	0.72	0.32	0.31	0.32
Islamophobia	0.70	0.32	0.62	0.42
Islamophobia	0.67	0.30	0.86	0.44
Anti-Black	0.82	0.97	0.38	0.55
Anti-Black	0.76	0.86	0.67	0.76
Anti-Black	0.73	0.68	0.81	0.74
Antisemitism	0.76	0.59	0.31	0.41
Antisemitism	0.72	0.55	0.59	0.57
Antisemitism	0.69	0.50	0.79	0.61
Anti-LGBTQ+	0.84	0.62	0.24	0.35
Anti-LGBTQ+	0.74	0.72	0.57	0.64
Anti-LGBTQ+	0.70	0.70	0.83	0.76

Table A2. Overall Antisemitic Metrics (Soft Score) by Threshold.

Threshold	Recall	Precision	F_1 Score
0.7	0.51	0.50	0.51
0.6	0.94	0.38	0.54
0.5	1.00	0.26	0.41